UMC 0.11um RFCMOS Al Advanced Enhancement High Speed 1.2V SPICE Model Document

Version 0.3 phase2

2013/04/15

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1 Introduction

1.1 Revision history

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2010/07/28	Yan Zung	1. Original release 2. Silicon-based model
0.3_p1	2010/12/20	Yan Zung	1. Add the models of noise figure, gate current and triple well NMOS. (Triple-well NMOS model was copied from twin well NMOS model)
0.3_p2	2013/04/15	YZ Wang	1. Modify the Verilog-A file.

1.2 General Information

Following Logic/MM model is used as the main body of the subckt RF MOSFET model and it would be fine tuned in order to improve Gm and Rout characteristics.

l110ae_hs12_v032_rf

More information about the model is specified below.

Model File Name:
Spectre: l110ae_hs12_v032_rf.lib.scs, l110ae_hs12_v032_rf.mdl.scs, l110ae_hs12_rf_s.va
Hspice: l110ae_hs12_v032_rf.lib, l110ae_hs12_v032_rf.mdl, l110ae_hs12_rf_h.va
Eldo: l110ae_hs12_v032_rf.lib.eldo, l110ae_hs12_v032_rf.mdl.eldo, l110ae_hs12_rf_e.va

Inline Sub-circuit Model Name & Node Definition:
Twin Well NMOS: n_12_hsrf (drain gate source body)
PMOS: p_12_hsrf (drain gate source body psub)
Triple Well NMOS: n_12bpw_hsrf (drain gate source body nwell psub)

Corner Section:
tt : Typical n-ch MOSFET and Typical p-ch MOSFET
ff : Fast n-ch MOSFET and Fast p-ch MOSFET
ss : Slow n-ch MOSFET and Slow p-ch MOSFET
fnsp : Fast n-ch MOSFET and Slow p-ch MOSFET
snfp : Slow n-ch MOSFET and Fast p-ch MOSFET

Input Parameters:
lf : gate finger length in m
wf : gate finger width in m
nf : gate finger number
hfn_flag: local flag parameter for High-Frequency noise model, Default equals the global high-frequency noise flag parameter - hfnoise (Please refer to the chapter of "High Frequency Noise Model Enhancement" for detail)
mf: multiple factor for high frequency noise model (for Hspice and Eldo only). mf should be always equal to the instant multiple factor parameter "m", in order to get accurate high frequency noise simulation results.

Geometry scaling:
NMOS:
Finger Length (lf): 0.12um ~ 0.36um
Finger Width (wf): 0.9um ~ 7.2um
Finger No (nf): 2~ 22
Support even and odd finger number

PMOS:
Finger Length (lf): 0.12um ~ 0.36um
Finger Width (wf): 1.6um ~ 9.6um
Finger No (nf): 2 ~ 22
Support even and odd finger number

Triple-well NMOS:
Finger Length (lf): 0.12um ~ 0.36um
Finger Width (wf): 0.9um ~ 7.2um
Finger No (nf): 2~ 22
Support even and odd finger number

Biasing coverage:
Voltage: Upto Vdd=1.2V
Frequency: upto 20.1GHz

Temperature coverage:
-50 C ~ +125 C

Simulator Versions:

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.11_1.1

Note: Cadence claimed (on 2010/12/06) that a bug of thermal noise model in Spectre will be fixed in the upcoming release of MMSIM72_ISR16

Wafer Information:

Table 1-2 Wafer Information

Mask ID	BSW0FA
Lot ID	D4YAA
Wafer Number	4

1.3 Model Usage:

1.3.1 For Cadence Spectre

- Copy ModelFileName.lib.scs and ModelFileName.mdl.scs & l110ae_hs12_s.va into your simulation directory.
- define global high frequency noise model parameter (hfnoise: 1 or 0) in the input netlist (Please refer to the chapter of "High Frequency Noise Model Enhancement" for detail)
parameters hfnoise=0
- Add the following statement in the input netlist to the simulator
include "./ModelFileName.lib.scs " section = CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
- SetOption1 options scale=0.9
-Follow the example below to call the model
x1 (drain gate source body) n_12_hsrf lf=0.12e-6 wf=3.6e-6 nf=4 m=8
+hfn_flag=1

1.3.2 For Synopsis Hspice

- Copy ModelFileName.lib and ModelFileName.mdl & l110ae_hs12_h.va into your simulation directory.
- define global high frequency noise model parameter (hfnoise: 1 or 0) in the input netlist (Please refer to the chapter of "High Frequency Noise Model Enhancement" for detail)
.param hfnoise=0
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
-Follow the example below to call the model
x1 drain gate source body n_12_hsrf lf=0.12e-6 wf=3.6e-6 nf=4 m=8 mf=8
+hfn_flag=1

1.3.3 For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo and ModelFileName.mdl.eldo & l110ae_hs12_e.va into your simulation directory.
- define global high frequency noise model parameter (hfnoise: 1 or 0) in the input netlist (Please refer to the chapter of "High Frequency Noise Model Enhancement" for detail)
.param hfnoise=0
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9

- Follow the example below to call the model
x1 drain gate source body n_12_hsrf lf=0.12e-6 wf=3.6e-6 nf=4 m=8 mf=8
+hfn_flag=1

ModelFileName: l110ae_hs12_v032_rf

1.4 Device Architecture and Layout

1.4.1 Twin Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

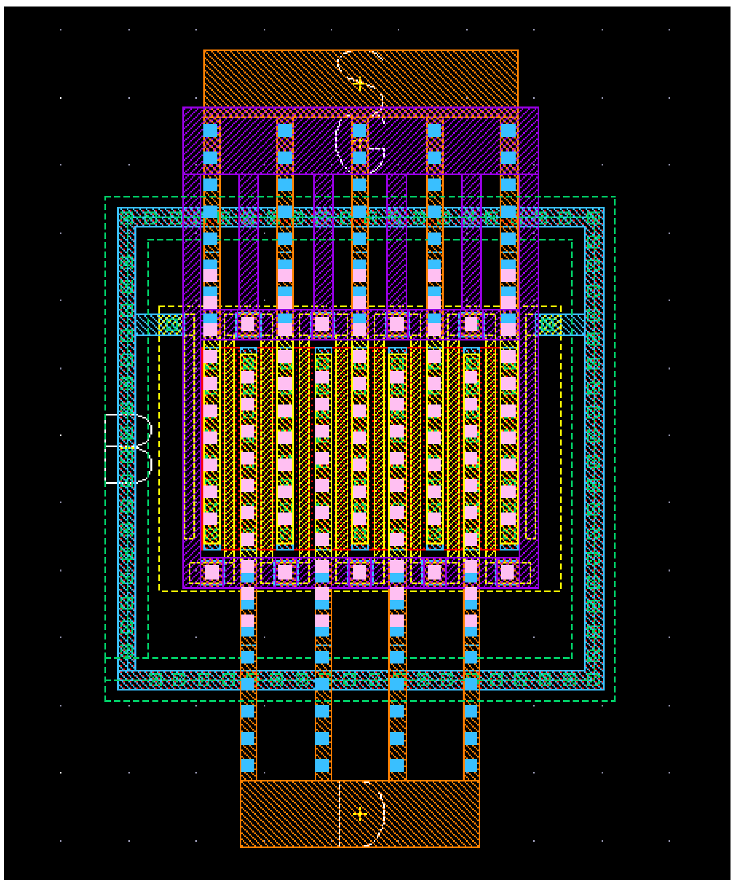


Figure 1-1 Example layout of 1.2V pwell RF NMOS

Extension equivalent circuit model:

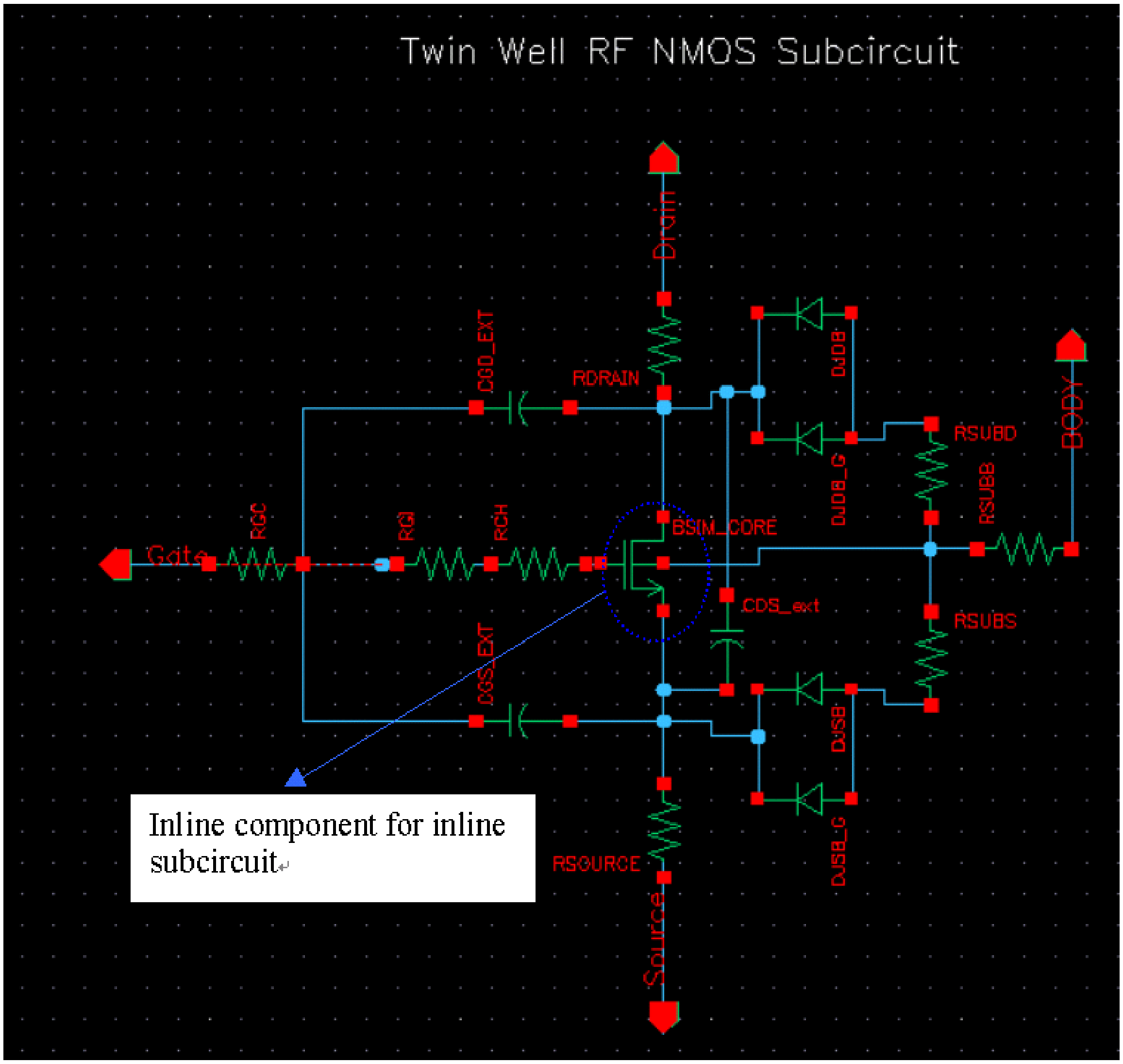


Figure 1-2 Extension equivalent circuit of TWIN-Well NMOS

1.4.2 PMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

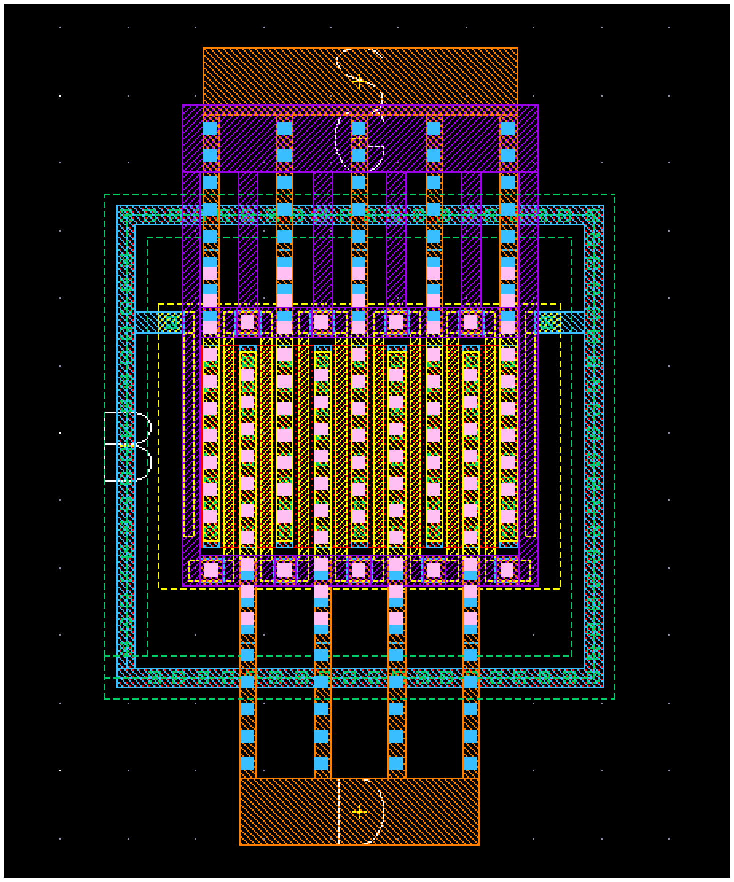


Figure 1-3 Example layout of 1.2V pwell RF NMOS

Extension equivalent circuit model:

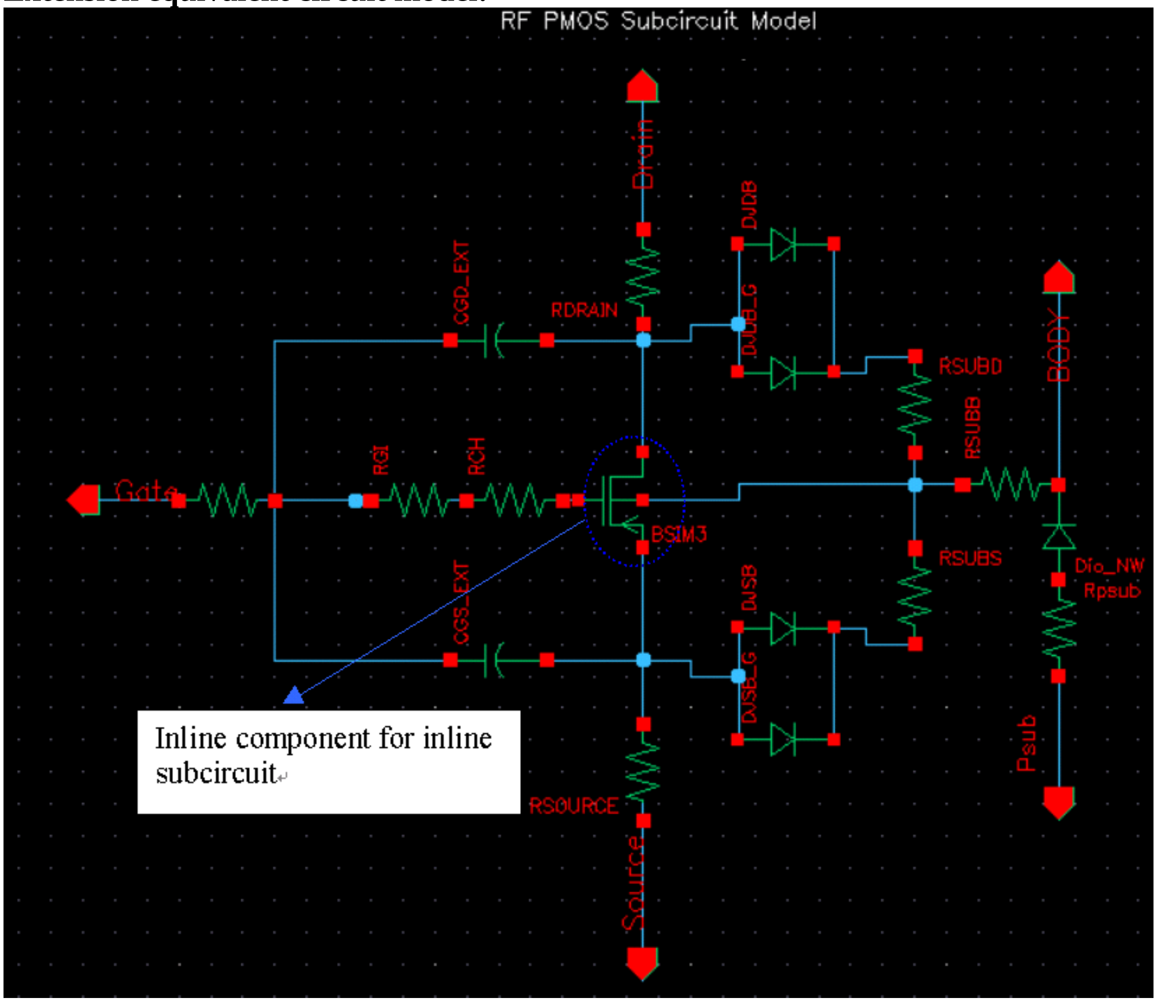


Figure 1-4 Extension equivalent circuit of PMOS

1.4.3 Triple Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

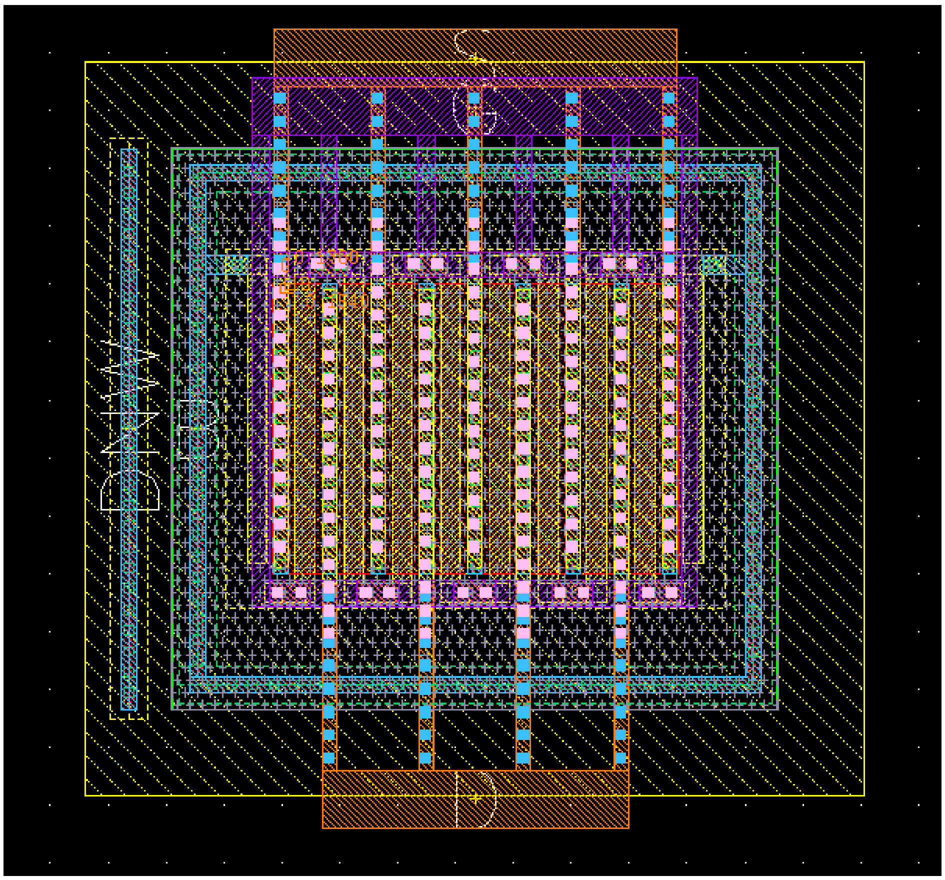


Figure 1-5 Example layout of Triple-Well RF NMOS

Extension equivalent circuit model:

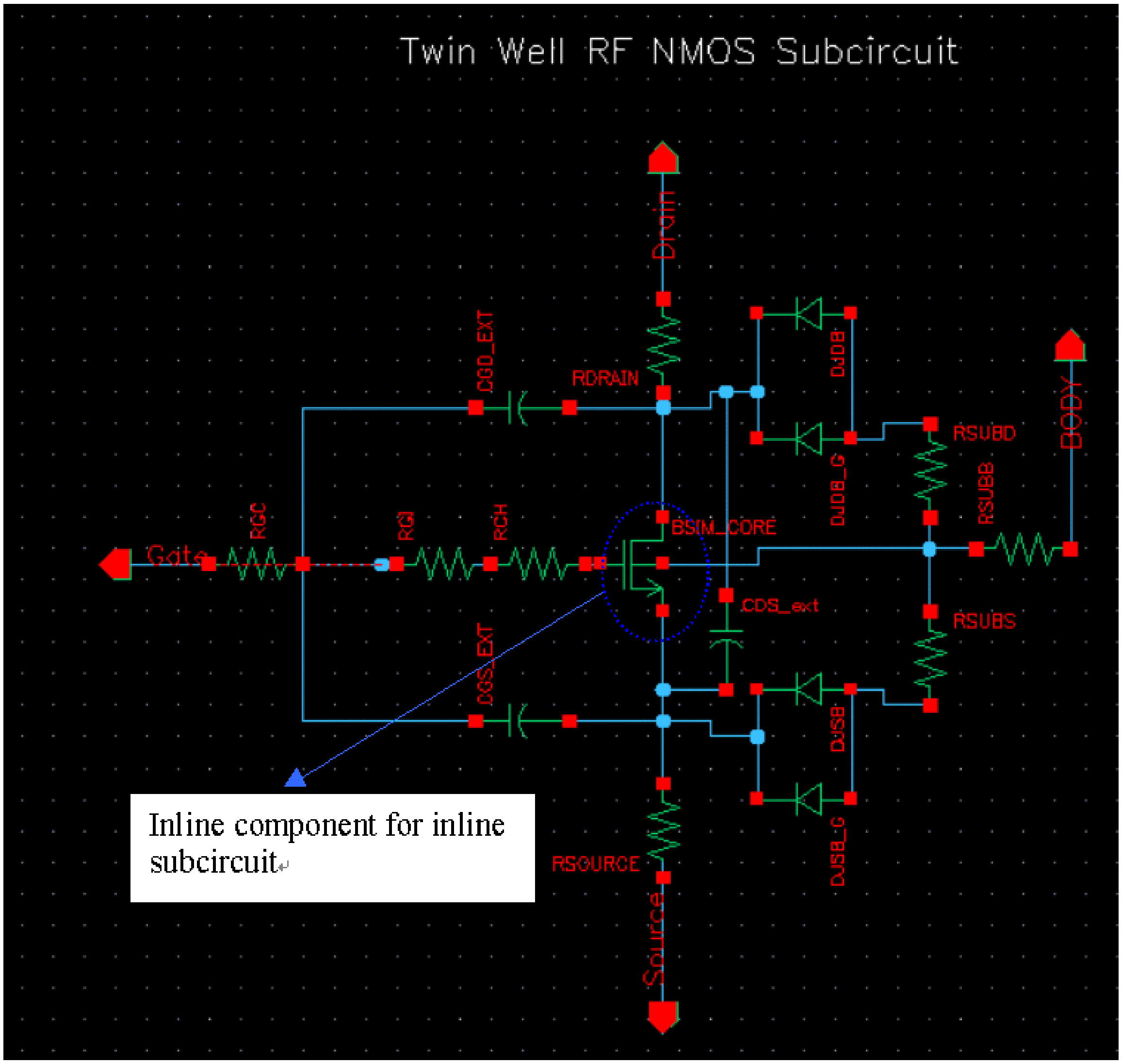


Figure 1-6 Sub-circuit topology of 1.2V Triple-Well RF NMOS

1.5 Model Features

BSIM3 :
The BSIM3model in RF subcircuit model is based on Logic/MM model, with parameters refined to improve the fitting quality of Gm/ Rout and RF performance.

LOD effect :
LOD model has been added to model STI stress & Finger number effects.
1/f Noise :
The BSIM3 1/f noise model is included

Note:
Model can support odd finger numbers, but it is recommended to use even finger number to reduce drain junction capacitance.
It is recommended not to use the model outside the above WF, LF and NF range. Outside above range, the model fitting quality may become worse.
This model should be used in conjunction with RF FET Pcell (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The model may not be valid if the transistor layout is different from UMC Pcell as shown in above layout example.

2 RF Model Verification

2.1 Twin Well NMOS Model

2.1.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.
The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max : Vd=1.2V, sweep Vg
vth : Linear Vth extracted @ Vd=0.05V @
Id=300nA*WF*0.9*NF/(LF*0.9+0.008um)
vtsat : Saturation Vth extracted @ Vd=1.2V using fixed current
Id=300nA*WF*0.9*NF/(LF*0.9+0.008um)
body effect : vtsat shift for Vbody =0 ~ -1.2V
Idsat : Drain current extracted @ Vd=Vg=1.2V
Idlin : Drain current extracted @ Vd=0.05V, Vg=1.2V
Ioff : Drain current extracted @ Vg=0V, Vd=1.2V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	0.36	0.12	0.12	0.12	0.36	0.12	0.12	0.12
	WF (um)	3.6	7.2	1.8	0.9	3.6	7.2	1.8	0.9
	NF	16	16	2	16	16	16	2	16
SS	ft(GHz)	22.158	86.393	67.451	64.703	NA	NA	NA	NA
	idsat(uA/um)	2.80E+02	5.12E+02	5.37E+02	5.46E+02	2.86E+02	5.71E+02	5.62E+02	5.61E+02
	vtlin(V)	3.49E-01	4.59E-01	4.70E-01	4.68E-01	3.54E-01	4.32E-01	4.37E-01	4.37E-01
	vtsat(V)	3.15E-01	3.19E-01	3.35E-01	3.47E-01	3.20E-01	3.28E-01	3.33E-01	3.32E-01
	body_effect(V)	1.33E-01	4.80E-02	5.98E-02	6.45E-02	1.20E-01	3.62E-02	4.91E-02	6.58E-02
	ioff(pA/um)	6.50E+01	7.46E+02	4.50E+02	3.16E+02	4.37E+01	3.35E+02	2.65E+02	2.47E+02
SNFP	ft(GHz)	23.207	94.909	74.659	72.299	NA	NA	NA	NA
	idsat(uA/um)	3.04E+02	5.56E+02	5.90E+02	6.10E+02	3.11E+02	6.25E+02	6.21E+02	6.27E+02
	vtlin(V)	3.23E-01	4.33E-01	4.37E-01	4.27E-01	3.26E-01	3.96E-01	3.98E-01	3.94E-01
	vtsat(V)	2.89E-01	2.92E-01	3.01E-01	3.06E-01	2.92E-01	2.92E-01	2.93E-01	2.88E-01
	body_effect(V)	1.33E-01	5.12E-02	6.25E-02	6.64E-02	1.18E-01	3.31E-02	4.66E-02	6.35E-02
	ioff(pA/um)	1.45E+02	1.48E+03	1.10E+03	9.64E+02	1.05E+02	9.81E+02	8.71E+02	9.32E+02
TT	ft(GHz)	24.969	108.91	84.781	80.821	NA	NA	NA	NA
	idsat(uA/um)	3.29E+02	6.04E+02	6.46E+02	6.74E+02	3.36E+02	6.78E+02	6.79E+02	6.93E+02
	vtlin(V)	2.95E-01	3.94E-01	3.98E-01	3.86E-01	2.99E-01	3.61E-01	3.59E-01	3.51E-01
	vtsat(V)	2.61E-01	2.54E-01	2.60E-01	2.62E-01	2.65E-01	2.55E-01	2.53E-01	2.44E-01
	body_effect(V)	1.33E-01	3.54E-02	5.21E-02	6.14E-02	1.18E-01	3.15E-02	4.55E-02	6.32E-02
	ioff(pA/um)	3.67E+02	4.69E+03	3.77E+03	3.57E+03	2.56E+02	3.05E+03	3.03E+03	3.71E+03
FNSP	ft(GHz)	26.769	122.71	94.697	89.017	NA	NA	NA	NA
	idsat(uA/um)	3.55E+02	6.50E+02	7.03E+02	7.40E+02	3.61E+02	7.31E+02	7.38E+02	7.59E+02
	vtlin(V)	2.73E-01	3.60E-01	3.63E-01	3.46E-01	2.72E-01	3.25E-01	3.21E-01	3.08E-01
	vtsat(V)	2.40E-01	2.22E-01	2.23E-01	2.19E-01	2.38E-01	2.17E-01	2.12E-01	2.00E-01
	body_effect(V)	1.33E-01	1.49E-02	3.97E-02	5.58E-02	1.17E-01	2.98E-02	4.45E-02	6.29E-02
	ioff(pA/um)	7.48E+02	1.21E+04	1.13E+04	1.22E+04	6.32E+02	9.41E+03	1.04E+04	1.44E+04
FF	ft(GHz)	27.343	131.41	102.54	97.303	NA	NA	NA	NA
	idsat(uA/um)	3.80E+02	6.97E+02	7.60E+02	8.05E+02	3.87E+02	7.84E+02	7.98E+02	8.26E+02
	vtlin(V)	2.39E-01	3.23E-01	3.21E-01	3.00E-01	2.45E-01	2.90E-01	2.82E-01	2.66E-01
	vtsat(V)	2.06E-01	1.86E-01	1.82E-01	1.74E-01	2.11E-01	1.84E-01	1.76E-01	1.60E-01
	body_effect(V)	1.33E-01	1.92E-02	4.31E-02	5.82E-02	1.15E-01	2.93E-02	4.34E-02	6.05E-02
	ioff(pA/um)	2.23E+03	3.05E+04	3.27E+04	4.07E+04	1.50E+03	2.41E+04	2.94E+04	4.53E+04

Table 2-1 RF & Logic Corner Table at 25 ℃

Temp=125C		RF Model				Logic Model			
Device	LF (um)	0.36	0.12	0.12	0.12	0.36	0.12	0.12	0.12
	WF (um)	3.6	7.2	1.8	0.9	3.6	7.2	1.8	0.9
	NF	16	16	2	16	16	16	2	16
SS	ft(GHz)	17.01	72.293	55.47	52.851	NA	NA	NA	NA
	idsat(uA/um)	2.20E+02	4.54E+02	4.66E+02	4.68E+02	2.26E+02	4.96E+02	4.89E+02	4.89E+02
	vtlin(V)	2.91E-01	4.03E-01	4.12E-01	4.08E-01	3.00E-01	3.66E-01	3.71E-01	3.71E-01
	vtsat(V)	2.46E-01	2.52E-01	2.67E-01	2.78E-01	2.59E-01	2.60E-01	2.64E-01	2.63E-01
	body_effect(V)	1.38E-01	5.32E-02	6.41E-02	6.74E-02	1.24E-01	4.56E-02	5.78E-02	7.32E-02
	ioff(pA/um)	3.44E+03	2.06E+04	1.44E+04	1.11E+04	2.39E+03	1.24E+04	1.06E+04	1.03E+04
SNFP	ft(GHz)	17.882	79.296	61.288	58.913	NA	NA	NA	NA
	idsat(uA/um)	2.38E+02	4.91E+02	5.11E+02	5.22E+02	2.45E+02	5.43E+02	5.40E+02	5.46E+02
	vtlin(V)	2.64E-01	3.76E-01	3.78E-01	3.66E-01	2.72E-01	3.29E-01	3.31E-01	3.26E-01
	vtsat(V)	2.21E-01	2.24E-01	2.33E-01	2.36E-01	2.31E-01	2.23E-01	2.24E-01	2.18E-01
	body_effect(V)	1.38E-01	5.66E-02	6.70E-02	6.95E-02	1.22E-01	4.26E-02	5.51E-02	7.10E-02
	ioff(pA/um)	6.30E+03	3.48E+04	2.82E+04	2.58E+04	4.67E+03	2.78E+04	2.60E+04	2.80E+04
TT	ft(GHz)	19.203	91.178	69.402	65.329	NA	NA	NA	NA
	idsat(uA/um)	2.56E+02	5.36E+02	5.60E+02	5.75E+02	2.64E+02	5.89E+02	5.91E+02	6.03E+02
	vtlin(V)	2.35E-01	3.36E-01	3.38E-01	3.24E-01	2.44E-01	2.93E-01	2.91E-01	2.82E-01
	vtsat(V)	1.92E-01	1.86E-01	1.91E-01	1.91E-01	2.03E-01	1.85E-01	1.82E-01	1.73E-01
	body_effect(V)	1.39E-01	4.05E-02	5.65E-02	6.47E-02	1.22E-01	4.10E-02	5.42E-02	7.08E-02
	ioff(pA/um)	1.26E+04	8.03E+04	6.99E+04	6.80E+04	9.13E+03	6.42E+04	6.53E+04	7.75E+04
FNSP	ft(GHz)	20.547	103.23	77.466	71.484	NA	NA	NA	NA
	idsat(uA/um)	2.76E+02	5.80E+02	6.11E+02	6.31E+02	2.83E+02	6.36E+02	6.42E+02	6.60E+02
	vtlin(V)	2.12E-01	3.01E-01	3.02E-01	2.84E-01	2.16E-01	2.56E-01	2.51E-01	2.39E-01
	vtsat(V)	1.70E-01	1.53E-01	1.53E-01	1.47E-01	1.75E-01	1.47E-01	1.41E-01	1.28E-01
	body_effect(V)	1.38E-01	2.00E-02	4.41E-02	5.95E-02	1.22E-01	3.93E-02	5.32E-02	7.09E-02
	ioff(pA/um)	2.15E+04	1.59E+05	1.55E+05	1.68E+05	1.79E+04	1.47E+05	1.61E+05	2.08E+05
FF	ft(GHz)	20.977	110.34	83.662	77.872	NA	NA	NA	NA
	idsat(uA/um)	2.95E+02	6.21E+02	6.59E+02	6.84E+02	3.02E+02	6.81E+02	6.93E+02	7.18E+02
	vtlin(V)	1.78E-01	2.64E-01	2.60E-01	2.37E-01	1.89E-01	2.20E-01	2.12E-01	1.96E-01
	vtsat(V)	1.35E-01	1.16E-01	1.11E-01	1.01E-01	1.48E-01	1.13E-01	1.04E-01	8.72E-02
	body_effect(V)	1.39E-01	2.45E-02	4.77E-02	6.17E-02	1.20E-01	3.87E-02	5.19E-02	6.83E-02
	ioff(pA/um)	4.76E+04	3.17E+05	3.42E+05	4.06E+05	3.39E+04	2.93E+05	3.45E+05	4.79E+05

Table 2-2 RF & Logic Corner Table at 125 ℃

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	0.36	0.12	0.12	0.12	0.36	0.12	0.12	0.12
	WF (um)	3.6	7.2	1.8	0.9	3.6	7.2	1.8	0.9
	NF	16	16	2	16	16	16	2	16
SS	ft(GHz)	27.065	97.154	77.158	74.663	NA	NA	NA	NA
	idsat(uA/um)	3.42E+02	5.50E+02	5.87E+02	6.05E+02	3.51E+02	6.28E+02	6.17E+02	6.15E+02
	vtlin(V)	4.04E-01	5.16E-01	5.29E-01	5.28E-01	4.02E-01	4.94E-01	4.99E-01	4.99E-01
	vtsat(V)	3.74E-01	3.79E-01	3.98E-01	4.12E-01	3.71E-01	3.91E-01	3.96E-01	3.95E-01
	body_effect(V)	1.30E-01	4.46E-02	5.71E-02	6.24E-02	1.18E-01	2.84E-02	4.20E-02	5.99E-02
	ioff(pA/um)	1.19E+00	8.99E+00	6.50E+00	5.78E+00	9.37E-01	2.87E+00	3.24E+00	4.50E+00
SNFP	ft(GHz)	28.383	106.79	85.474	83.523	NA	NA	NA	NA
	idsat(uA/um)	3.71E+02	5.97E+02	6.47E+02	6.77E+02	3.82E+02	6.85E+02	6.80E+02	6.86E+02
	vtlin(V)	3.78E-01	4.90E-01	4.95E-01	4.87E-01	3.75E-01	4.59E-01	4.61E-01	4.56E-01
	vtsat(V)	3.49E-01	3.53E-01	3.64E-01	3.71E-01	3.43E-01	3.56E-01	3.57E-01	3.52E-01
	body_effect(V)	1.30E-01	4.81E-02	5.99E-02	6.42E-02	1.16E-01	2.53E-02	3.92E-02	5.76E-02
	ioff(pA/um)	2.01E+00	2.16E+01	1.61E+01	1.46E+01	1.37E+00	1.08E+01	1.00E+01	1.20E+01
TT	ft(GHz)	30.623	122.12	96.925	93.44	NA	NA	NA	NA
	idsat(uA/um)	4.02E+02	6.45E+02	7.05E+02	7.47E+02	4.13E+02	7.42E+02	7.43E+02	7.57E+02
	vtlin(V)	3.51E-01	4.52E-01	4.57E-01	4.47E-01	3.49E-01	4.24E-01	4.23E-01	4.14E-01
	vtsat(V)	3.21E-01	3.15E-01	3.24E-01	3.27E-01	3.17E-01	3.19E-01	3.17E-01	3.08E-01
	body_effect(V)	1.30E-01	3.19E-02	4.95E-02	5.94E-02	1.15E-01	2.33E-02	3.81E-02	5.69E-02
	ioff(pA/um)	4.20E+00	1.01E+02	7.44E+01	6.87E+01	2.81E+00	4.79E+01	4.72E+01	6.10E+01
FNSP	ft(GHz)	32.862	136.92	107.98	102.9	NA	NA	NA	NA
	idsat(uA/um)	4.32E+02	6.90E+02	7.63E+02	8.16E+02	4.44E+02	7.98E+02	8.06E+02	8.28E+02
	vtlin(V)	3.30E-01	4.18E-01	4.23E-01	4.09E-01	3.22E-01	3.90E-01	3.84E-01	3.72E-01
	vtsat(V)	3.00E-01	2.84E-01	2.87E-01	2.85E-01	2.90E-01	2.82E-01	2.77E-01	2.65E-01
	body_effect(V)	1.30E-01	1.15E-02	3.68E-02	5.40E-02	1.15E-01	2.15E-02	3.71E-02	5.68E-02
	ioff(pA/um)	9.02E+00	3.68E+02	3.20E+02	3.44E+02	7.65E+00	2.18E+02	2.44E+02	3.68E+02
FF	ft(GHz)	33.677	146.7	117.04	112.66	NA	NA	NA	NA
	idsat(uA/um)	4.64E+02	7.40E+02	8.25E+02	8.89E+02	4.77E+02	8.56E+02	8.70E+02	9.01E+02
	vtlin(V)	2.95E-01	3.82E-01	3.82E-01	3.63E-01	2.95E-01	3.54E-01	3.46E-01	3.31E-01
	vtsat(V)	2.66E-01	2.48E-01	2.46E-01	2.40E-01	2.63E-01	2.49E-01	2.41E-01	2.25E-01
	body_effect(V)	1.30E-01	1.57E-02	4.03E-02	5.58E-02	1.12E-01	2.12E-02	3.58E-02	5.45E-02
	ioff(pA/um)	3.67E+01	1.27E+03	1.35E+03	1.77E+03	2.26E+01	7.79E+02	9.98E+02	1.76E+03

Table 2-3 RF & Logic Corner Table at -50 ℃

2.1.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted Idsat is smaller than the real Idsat for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
NFET551	0.12	1.8	16	4	1	0.39	0.398	0.258	0.26	630.52	626.68	3874.04	3767.8	94.707	89.636
NFET552	0.12	2.4	16	3	1	0.391	0.399	0.257	0.259	624.17	613.07	4156.06	3909.8	100.348	93.749
NFET553	0.12	3.6	16	2	1	0.389	0.398	0.248	0.257	615.47	594.42	5462.19	4226.1	107.275	97.516
NFET554	0.12	7.2	16	1	1	0.389	0.394	0.246	0.254	565.43	561.59	5894.29	4691.5	105.67	99.751
NFET555	0.12	0.9	16	6	1	0.382	0.386	0.251	0.262	658.38	662.4	4751.03	3570.9	78.317	78.368
NFET556	0.12	1.8	2	18	1	0.397	0.398	0.269	0.26	630.59	638.39	2784.64	3768.5	83.716	82.721
NFET558	0.12	1.8	5	13	1	0.394	0.398	0.261	0.26	627.27	633.64	3576.26	3768	90.55	89.403
NFET559	0.12	1.8	8	8	1	0.389	0.398	0.258	0.26	631.66	631.32	3773.53	3767.9	91.034	89.912
NFET560	0.12	1.8	11	6	1	0.389	0.398	0.254	0.26	635.77	629.27	4416.39	3767.8	93.927	90.024
NFET561	0.12	1.8	22	3	1	0.391	0.398	0.256	0.26	634.23	625.57	4065.47	3767.8	94.465	89.659
NFET563	0.15	2.4	16	3	1	0.377	0.388	0.303	0.303	508.55	511.7	412.96	573.51	70.367	71.768
NFET564	0.18	2.4	16	3	1	0.347	0.36	0.295	0.299	460.63	465.03	334.86	403.98	55.236	57.998
NFET565	0.24	3.6	16	2	1	0.316	0.328	0.275	0.285	396.57	398.63	405.19	337.77	39.521	41.556
NFET566	0.36	3.6	16	2	1	0.291	0.295	0.255	0.261	315.9	321.67	445.14	367.21	22.907	24.39
NFET570	0.12	3.6	4	8	1	0.397	0.397	0.264	0.257	603.09	605.9	3647.18	4226.3	101.302	98.226
NFET571	0.12	3.6	11	4	1	0.393	0.398	0.256	0.257	596.07	596.73	4181.4	4226.1	104.213	97.377
NFET572	0.12	3.6	22	2	1	0.388	0.398	0.251	0.257	602.3	590.19	5086	4226.1	104.766	96.329
NFET573	0.15	7.2	4	4	1	0.386	0.395	0.313	0.306	467.13	487.81	307.12	512.07	74.087	76.618
NFET576	0.12	1.8	4	1	1	0.393	0.398	0.26	0.26	646.54	639.51	3610.49	3768.1	95.623	90.173

Table 2-4 Summary Table for Twin Well RF NMOS

Measure: Measurement result.
Model: Simulation result

2.1.3 Simulation results for WF, LF and NF trends

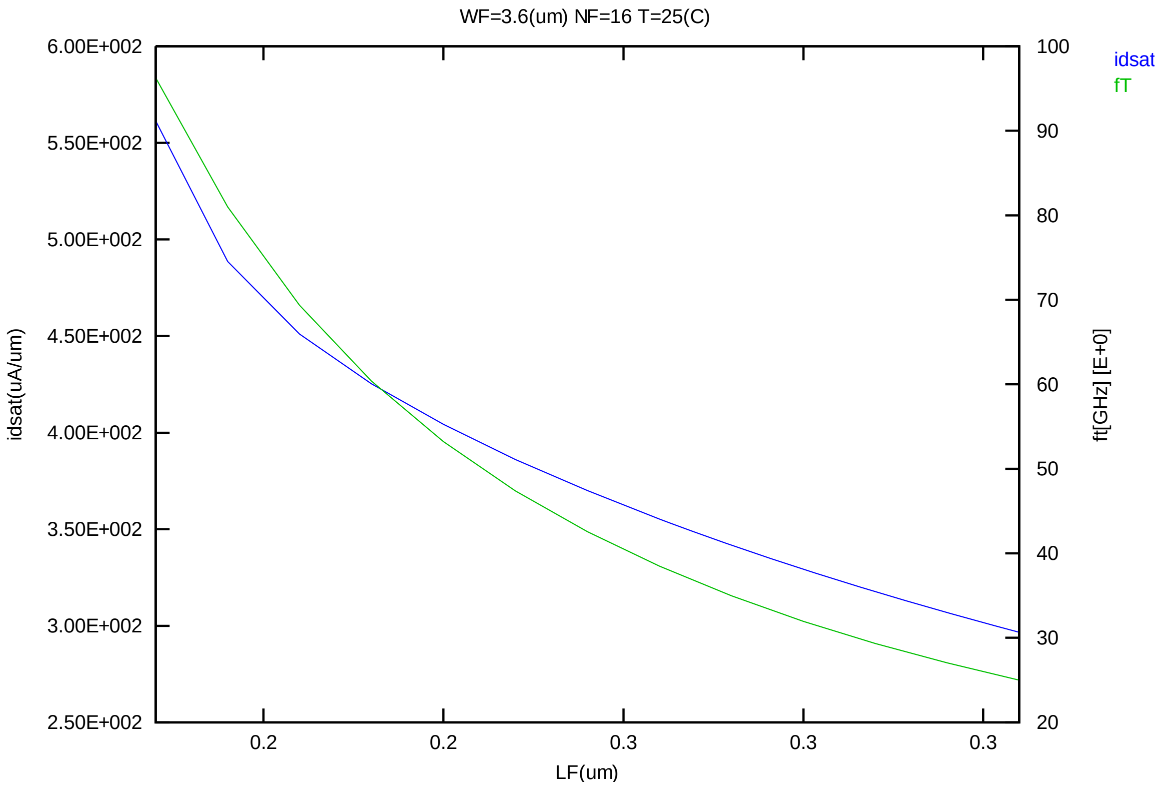


Figure 2-1 fT and Idsat vs. channel length

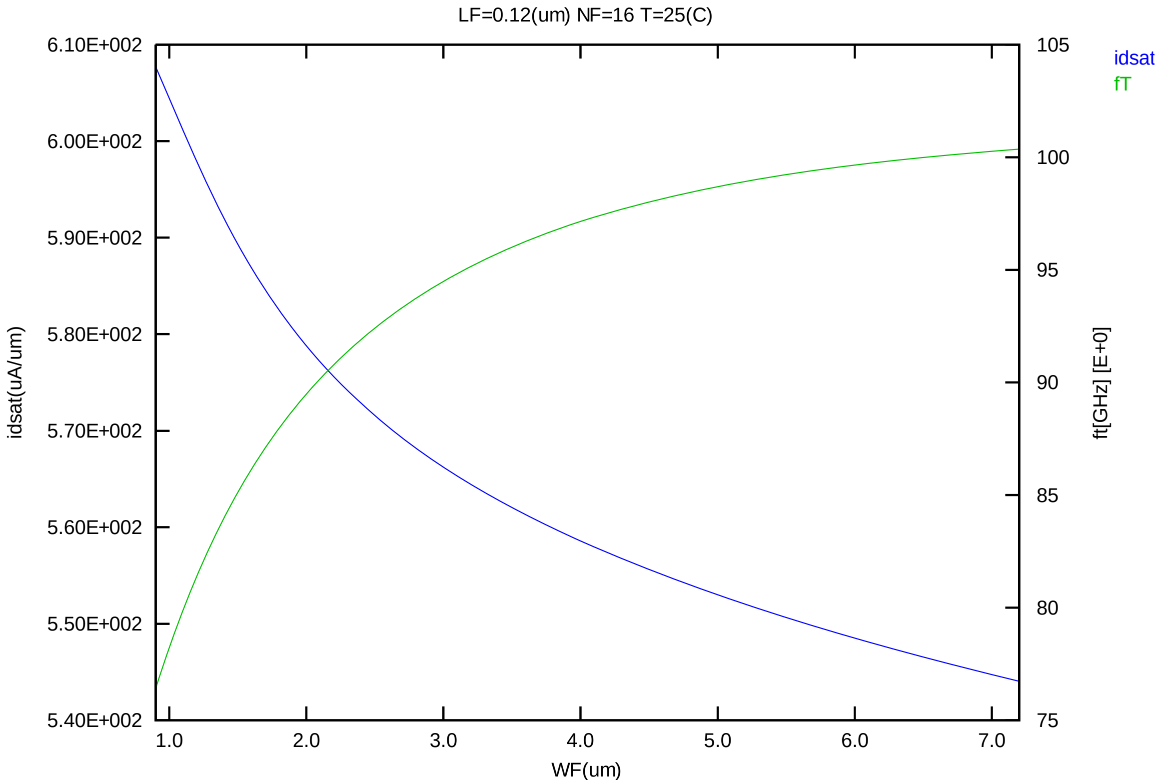


Figure 2-2 fT and Idsat vs. channel width

2.1.4 DC & RF Fitting curves for individual devices

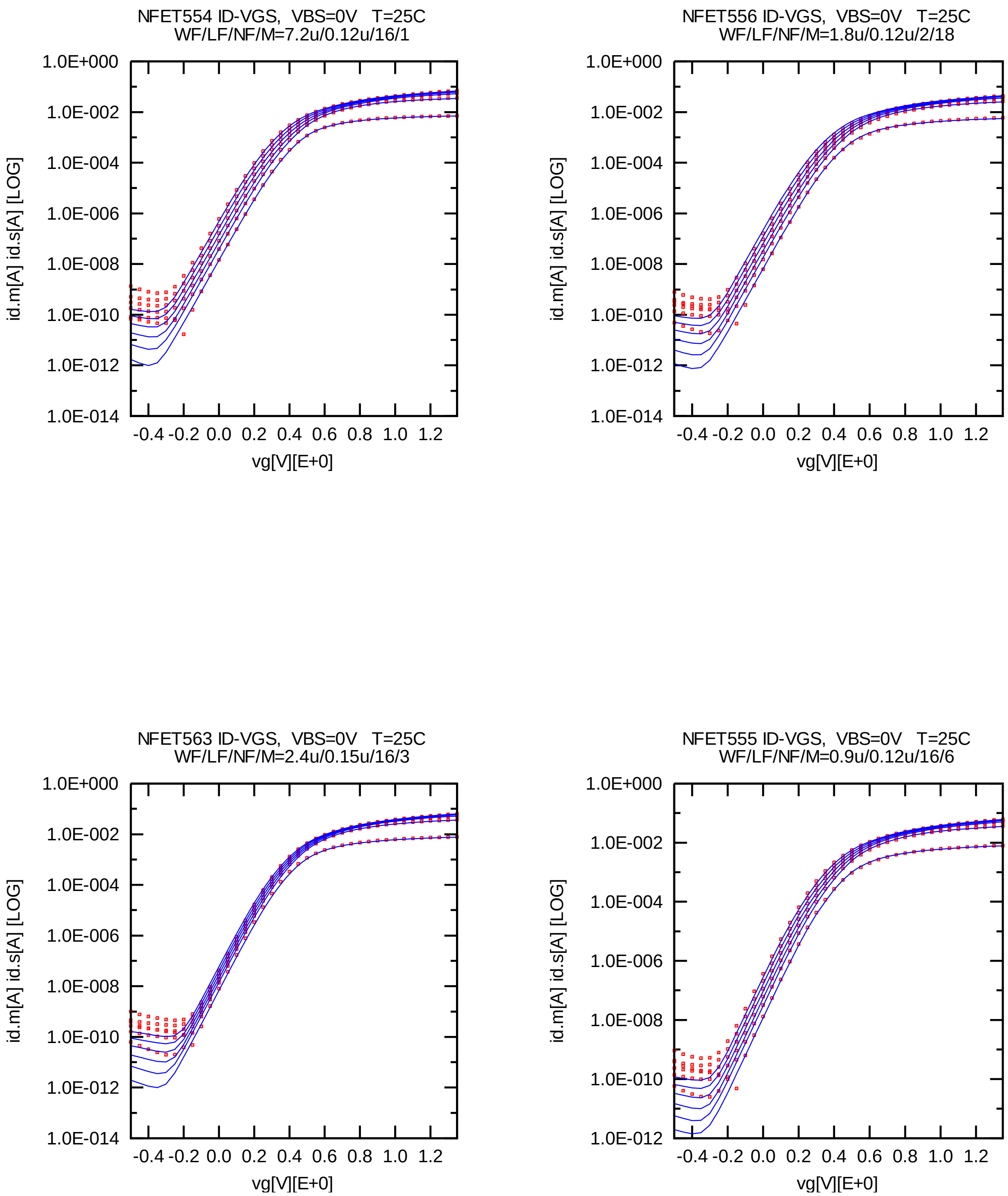


Figure 2-3 Id-Vg (25℃)

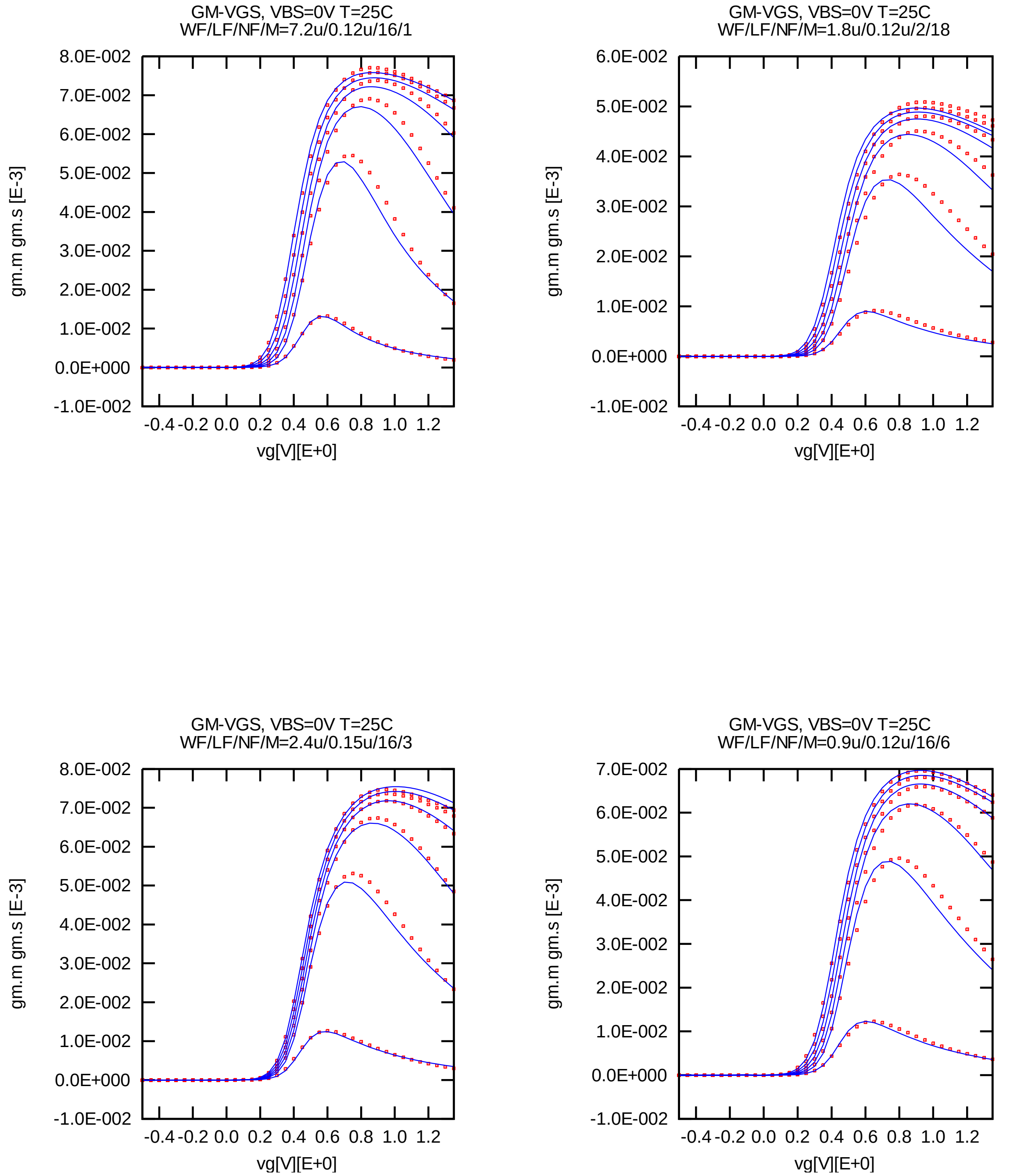


Figure 2-4 Gm-Vg (25℃)

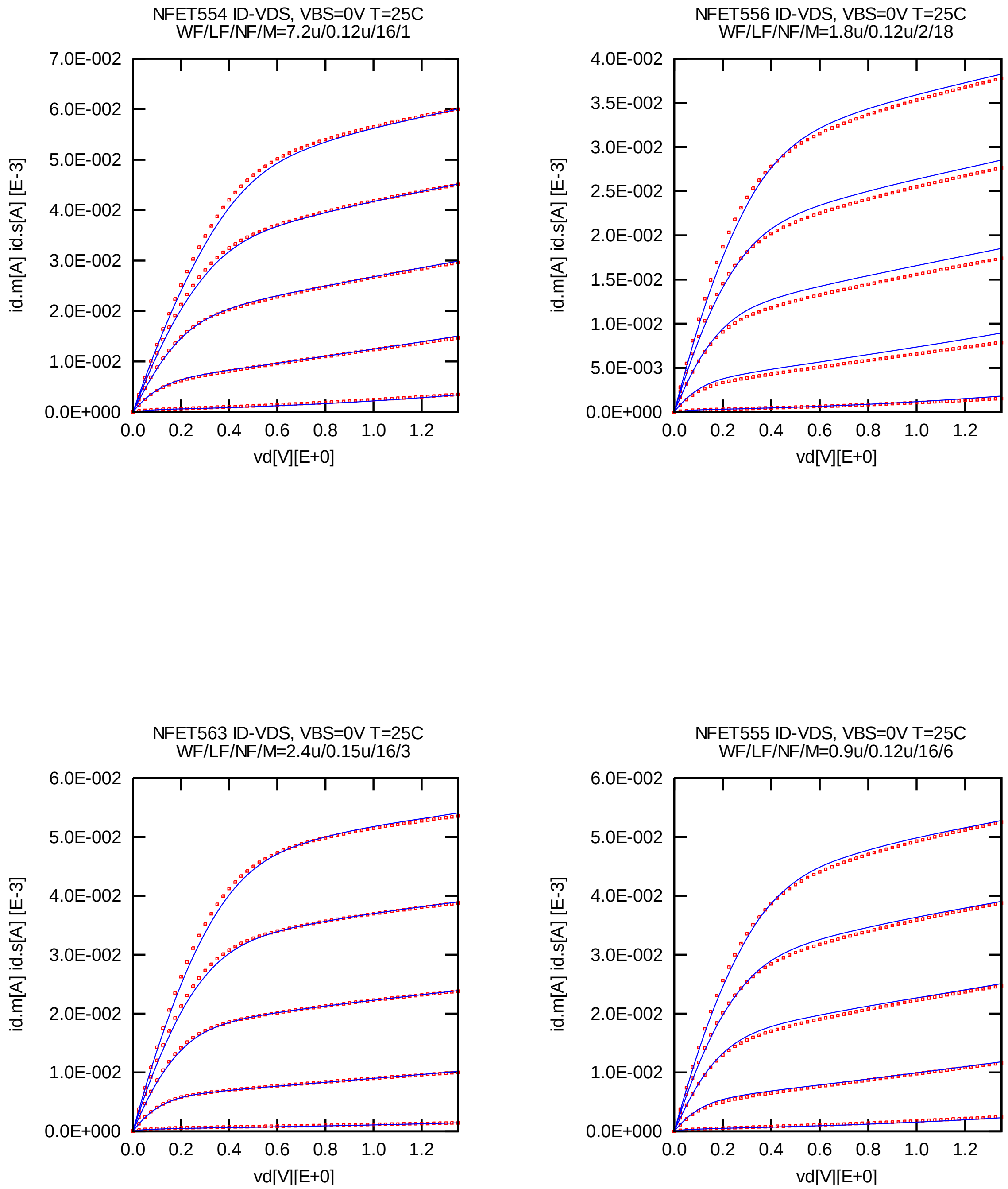


Figure 2-5 Id-Vd (25℃)

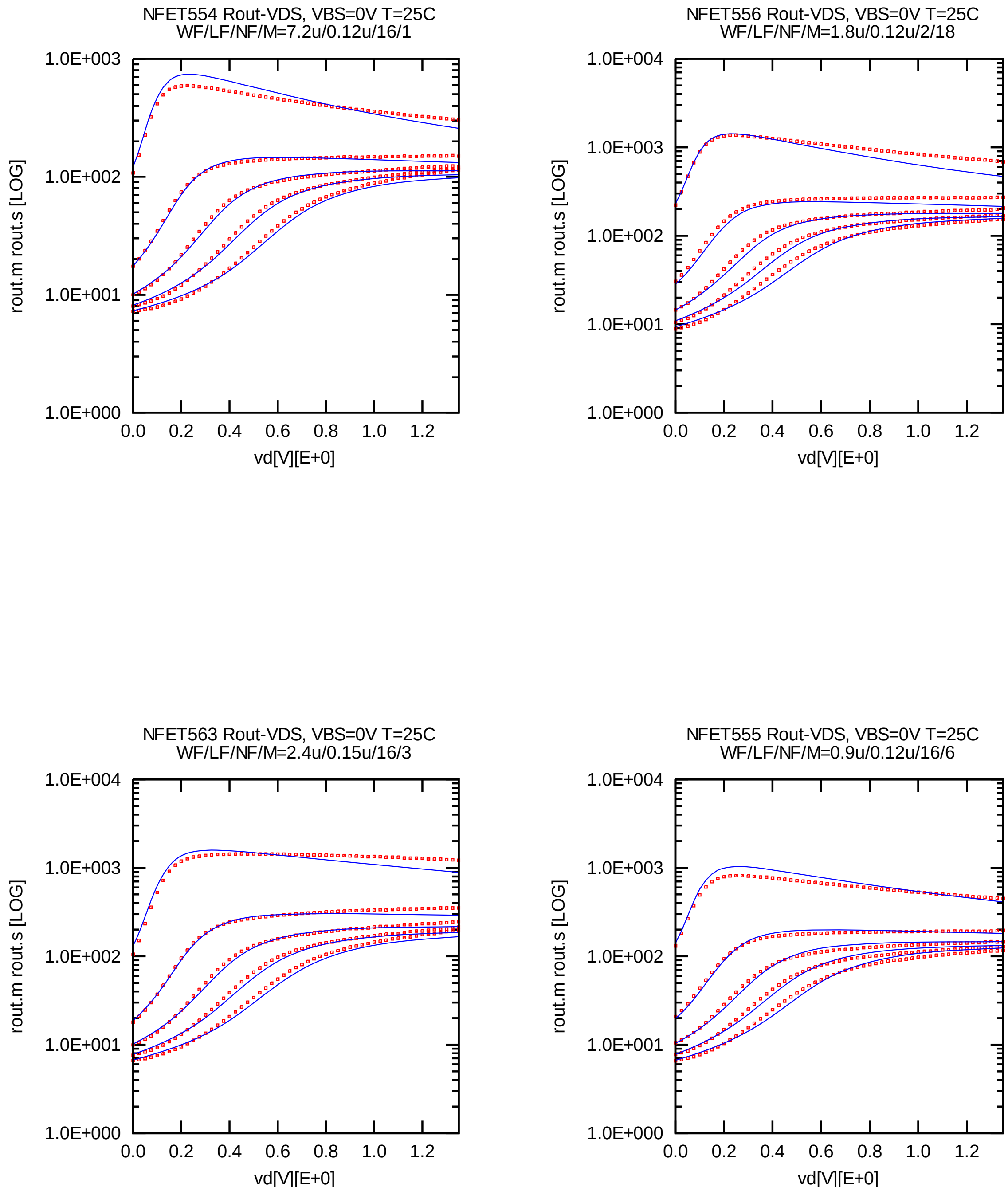


Figure 2-6 Rout-Vd (25℃)

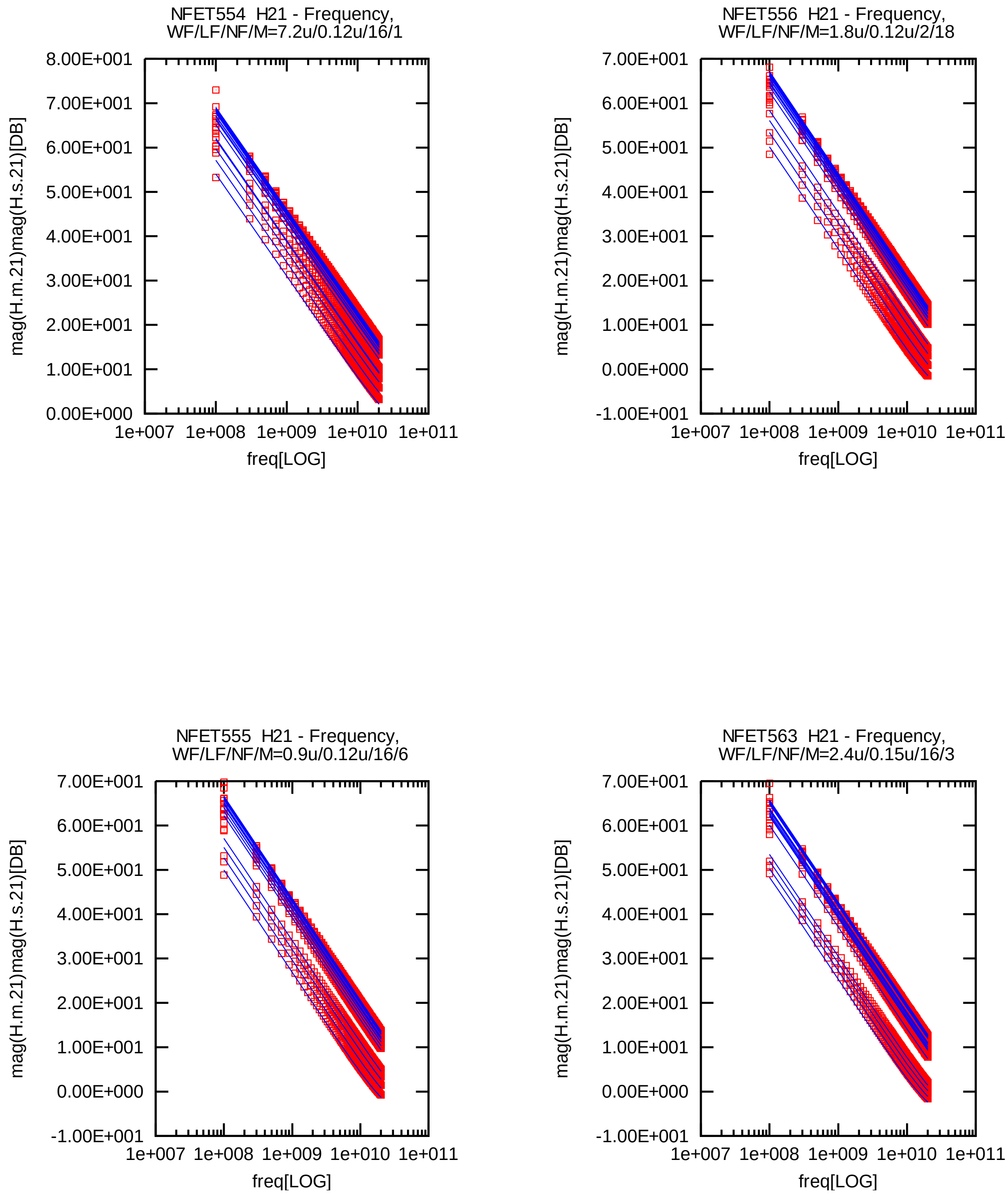


Figure 2-7 Mag(H21)-Freq (25℃)

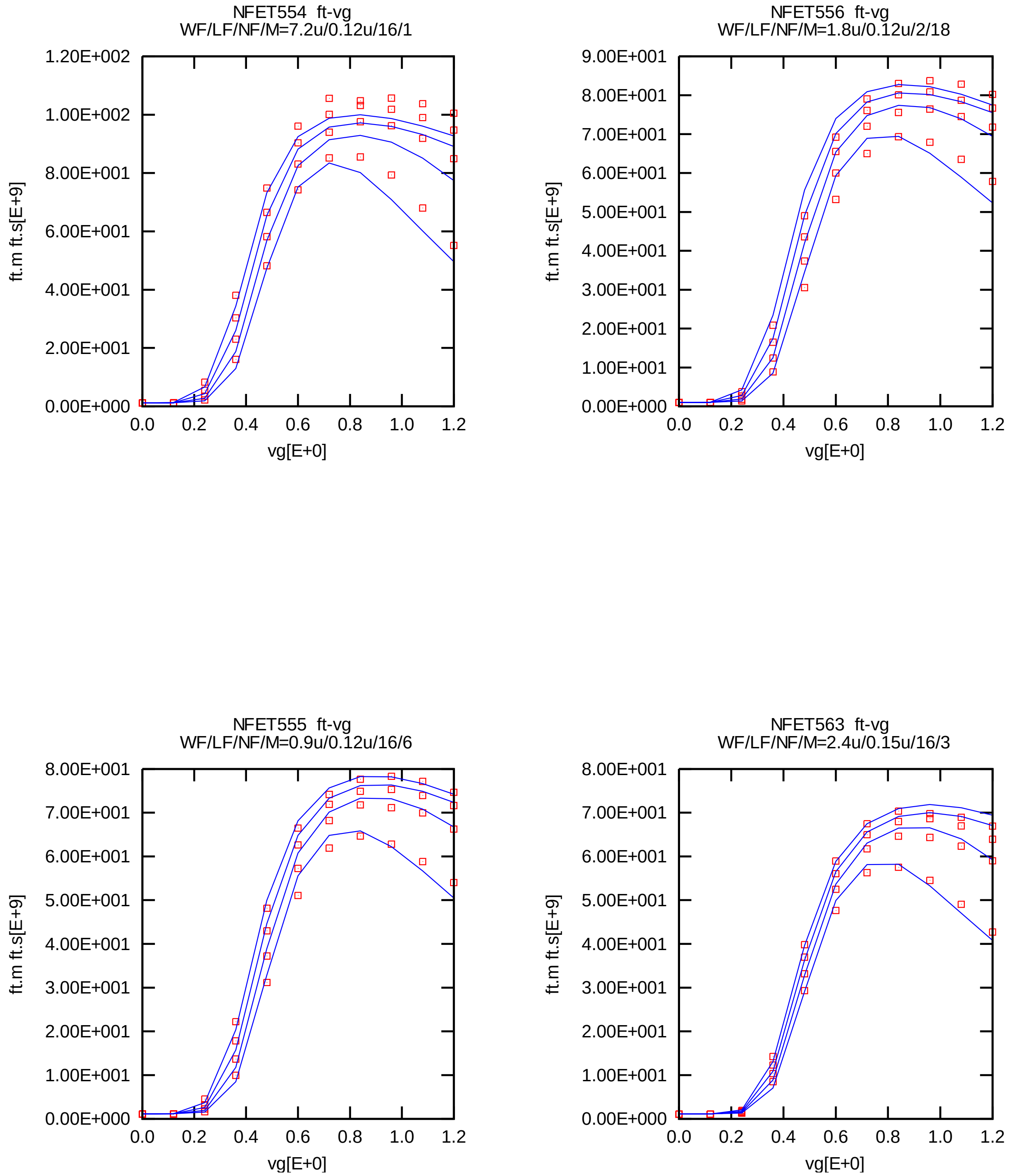


Figure 2-8 fT-Vg (25℃)

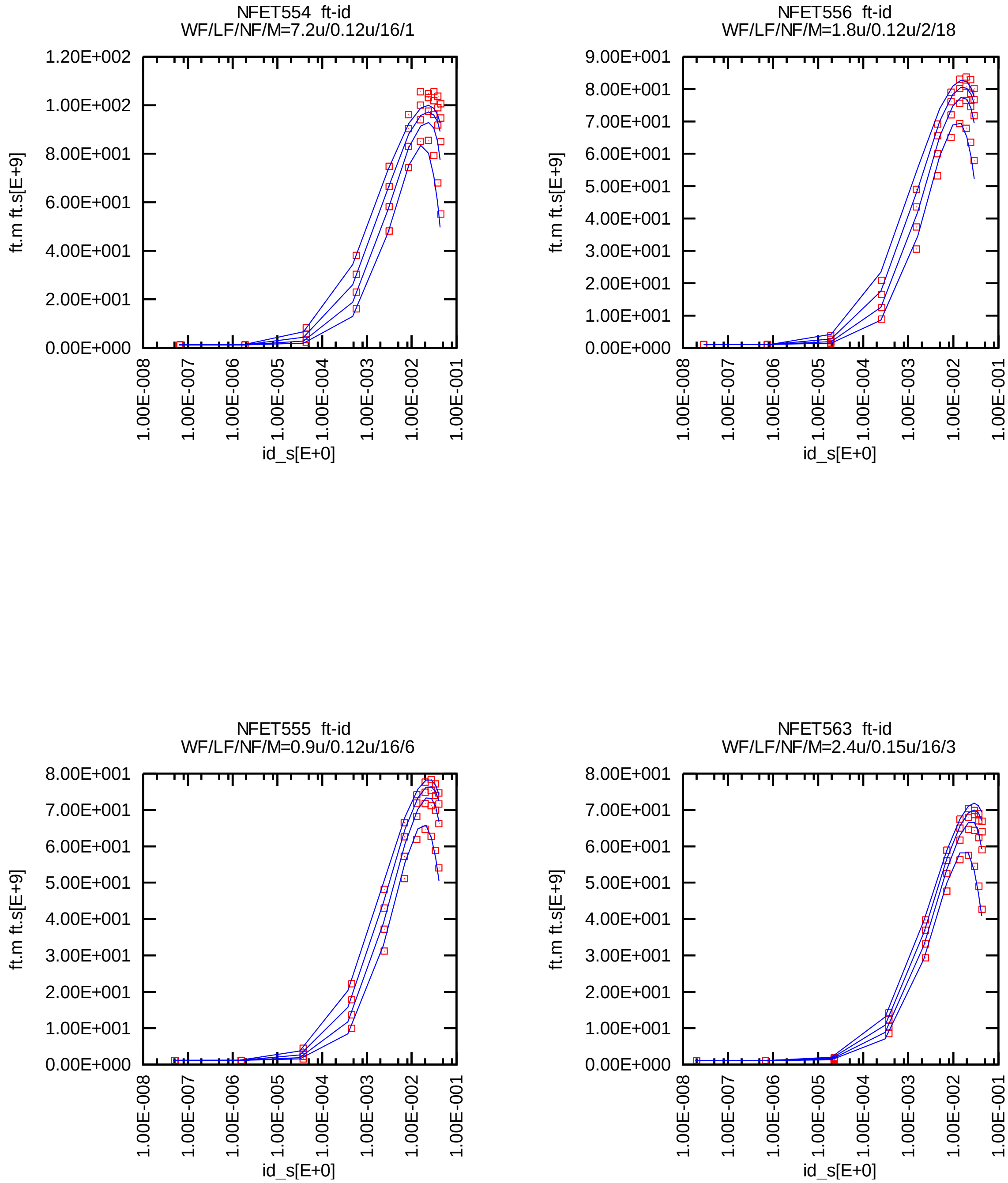


Figure 2-9 fT-Id (25℃)

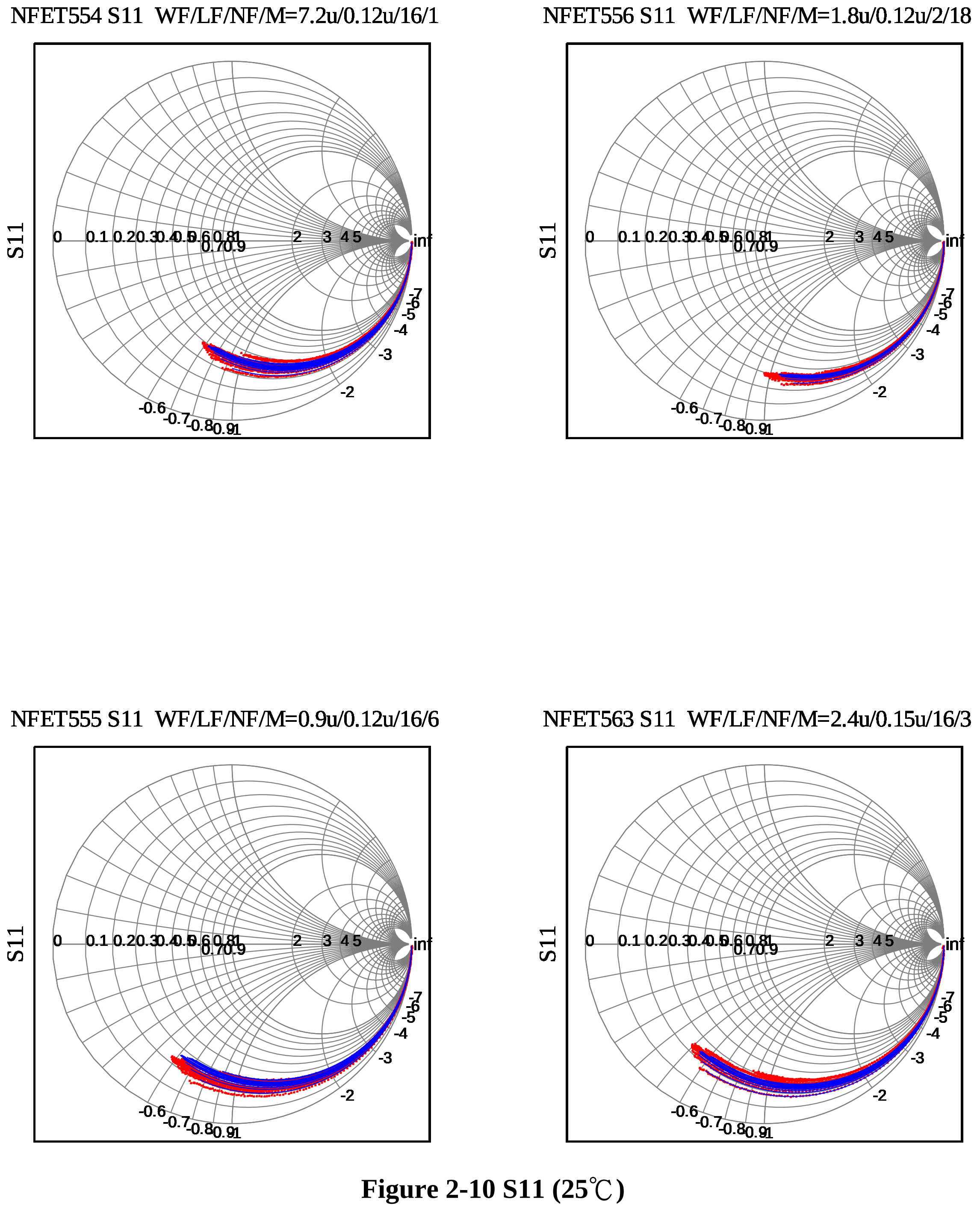


Figure 2-10 S11 (25℃)

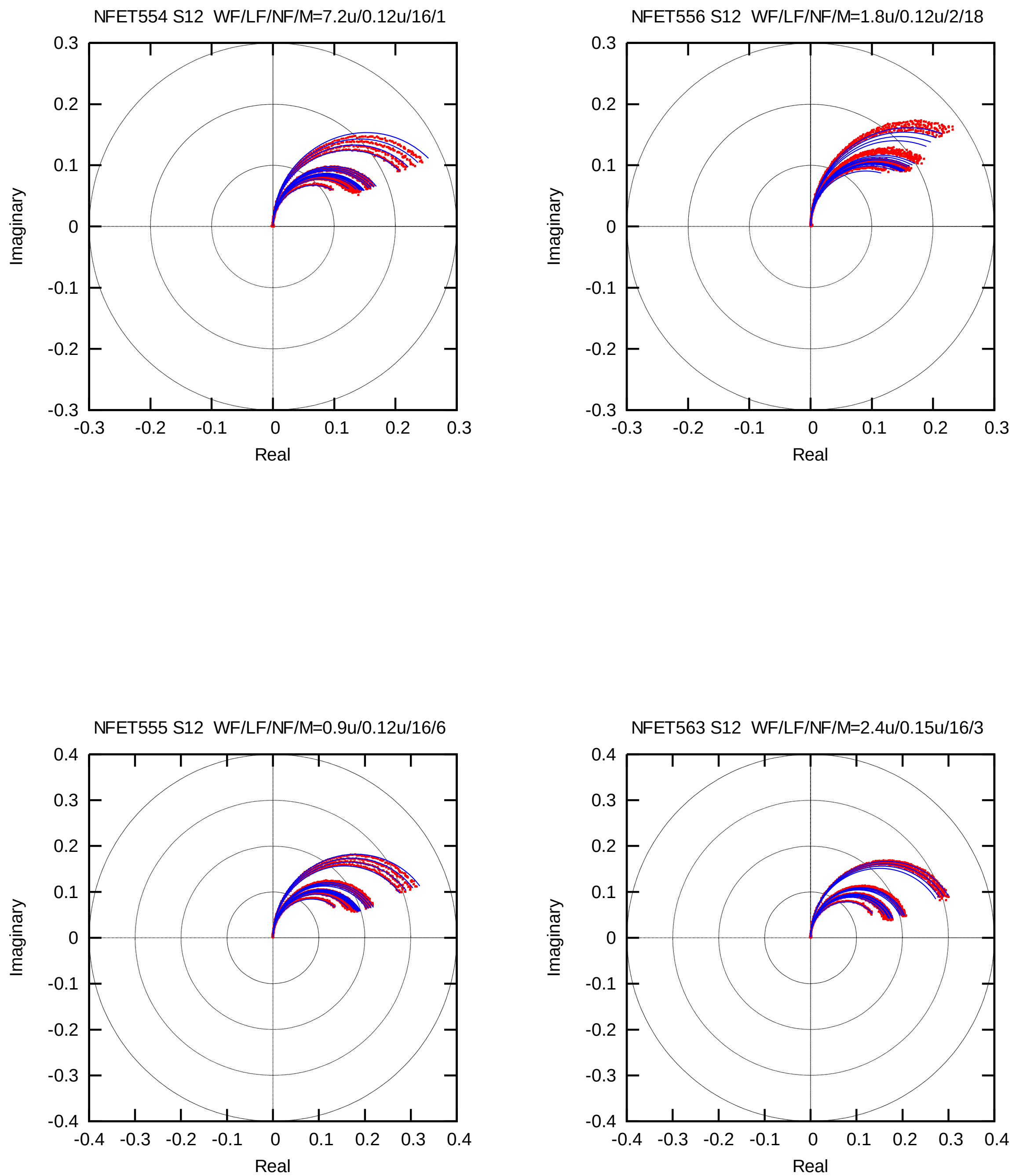


Figure 2-11 S12 (25℃)

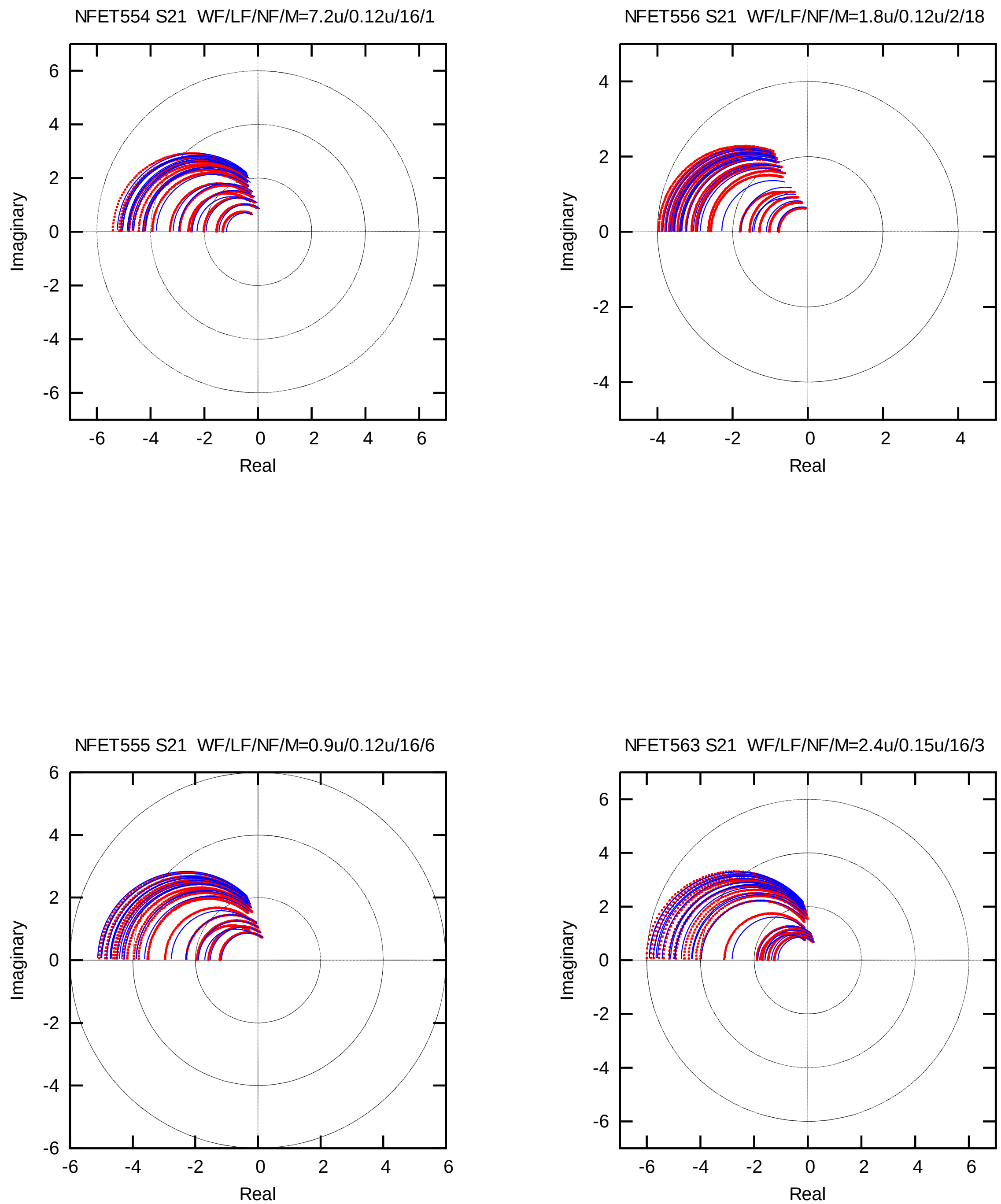


Figure 2-12 S21 (25℃)

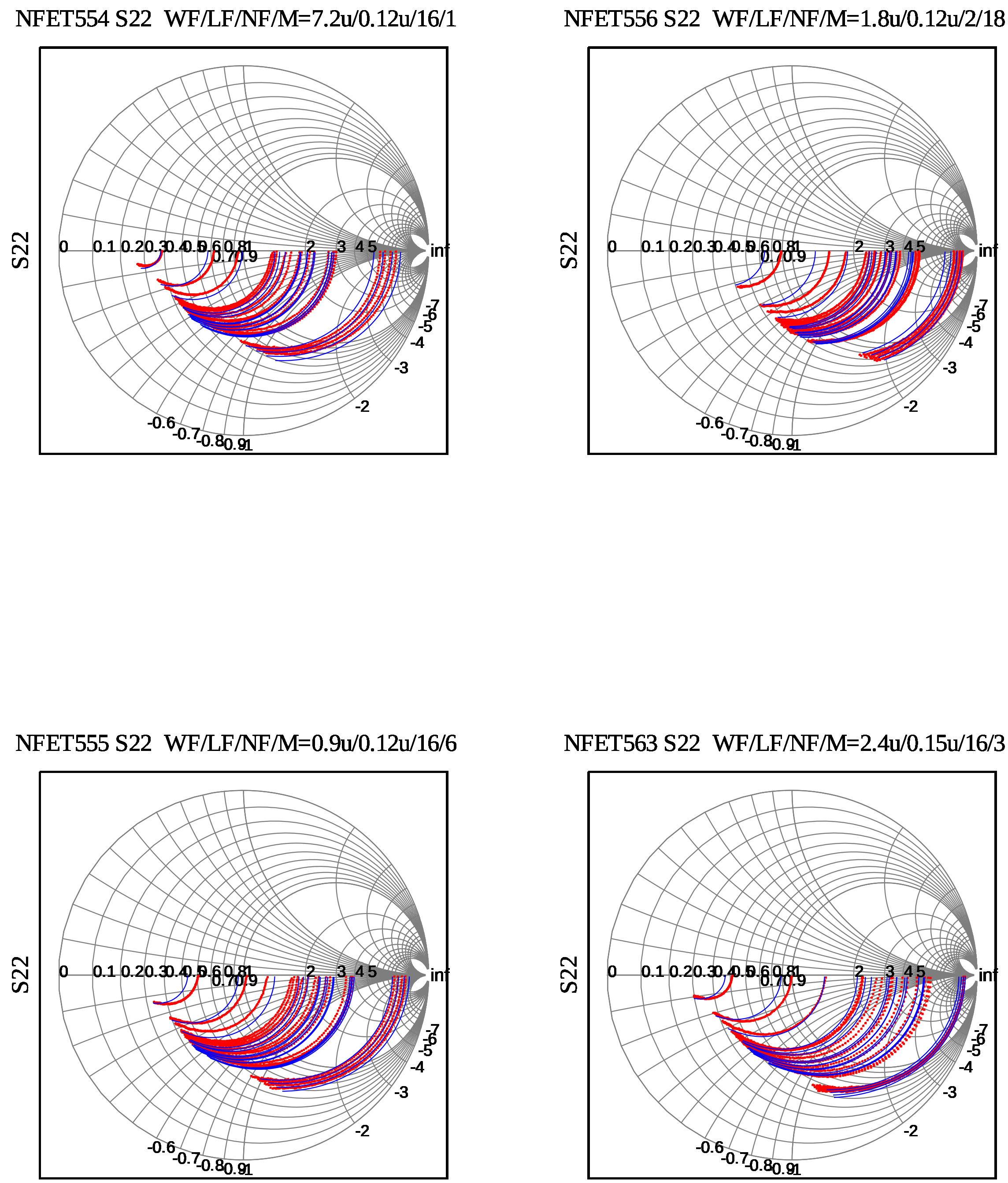


Figure 2-13 S22 (25℃)

2.2 PMOS Model

2.2.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.
The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max:Vd=-1.2V, sweep Vg
vth : Linear Vth extracted @ Vd=-0.05V @ Id=-70nA*WF*0.9*NF/(LF*0.9+0.008um)
vtsat : Saturation Vth extracted @ Vd=-1.2V using fixed current Id=-70nA*WF*0.9*NF/(LF*0.9+0.008um)
body effect : vtsat shift for Vbody =0 ~ 1.2V
Idsat : Drain current extracted @ Vd=Vg=-1.2V
Idlin : Drain current extracted @ Vd=-0.05, Vg=-1.2V
Ioff : Drain current extracted @ Vg=0V, Vd=-1.2V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	4	16	22	16	4	16
SS	ft(GHz)	14.575	45.878	41.198	35.561	NA	NA	NA	NA
	idsat(uA/um)	-1.08E+02	-2.26E+02	-2.30E+02	-2.23E+02	-1.04E+02	-2.31E+02	-2.23E+02	-2.18E+02
	vtlin(V)	-3.50E-01	-3.74E-01	-3.88E-01	-3.93E-01	-3.31E-01	-3.66E-01	-3.85E-01	-3.92E-01
	vtsat(V)	-3.16E-01	-2.72E-01	-2.88E-01	-2.93E-01	-2.97E-01	-2.68E-01	-2.89E-01	-2.97E-01
	body_effect(V)	-1.82E-01	-1.04E-01	-1.03E-01	-1.03E-01	-1.83E-01	-1.45E-01	-1.44E-01	-1.44E-01
	ioff(pA/um)	-3.63E+01	-5.89E+02	-4.10E+02	-3.86E+02	-8.29E+01	-7.99E+02	-4.35E+02	-3.32E+02
SNFP	ft(GHz)	18.352	64.826	57.965	49.87	NA	NA	NA	NA
	idsat(uA/um)	-1.41E+02	-3.18E+02	-3.33E+02	-3.25E+02	-1.40E+02	-3.23E+02	-3.25E+02	-3.24E+02
	vtlin(V)	-2.60E-01	-2.56E-01	-2.56E-01	-2.55E-01	-2.41E-01	-2.52E-01	-2.57E-01	-2.59E-01
	vtsat(V)	-2.26E-01	-1.41E-01	-1.44E-01	-1.43E-01	-2.07E-01	-1.47E-01	-1.53E-01	-1.56E-01
	body_effect(V)	-1.81E-01	-8.64E-02	-8.68E-02	-8.74E-02	-1.80E-01	-1.36E-01	-1.36E-01	-1.36E-01
	ioff(pA/um)	-5.83E+02	-2.03E+04	-1.97E+04	-2.16E+04	-1.21E+03	-1.87E+04	-1.54E+04	-1.40E+04
TT	ft(GHz)	17.006	58.018	52.285	45.218	NA	NA	NA	NA
	idsat(uA/um)	-1.30E+02	-2.86E+02	-2.98E+02	-2.91E+02	-1.28E+02	-2.92E+02	-2.92E+02	-2.89E+02
	vtlin(V)	-2.89E-01	-2.95E-01	-3.00E-01	-3.01E-01	-2.70E-01	-2.91E-01	-2.99E-01	-3.03E-01
	vtsat(V)	-2.55E-01	-1.85E-01	-1.92E-01	-1.93E-01	-2.37E-01	-1.86E-01	-1.97E-01	-2.02E-01
	body_effect(V)	-1.82E-01	-9.31E-02	-9.29E-02	-9.32E-02	-1.80E-01	-1.37E-01	-1.36E-01	-1.37E-01
	ioff(pA/um)	-2.34E+02	-6.26E+03	-5.50E+03	-5.78E+03	-4.97E+02	-6.84E+03	-4.93E+03	-4.23E+03
FNSP	ft(GHz)	15.689	51.837	46.725	40.463	NA	NA	NA	NA
	idsat(uA/um)	-1.19E+02	-2.55E+02	-2.63E+02	-2.56E+02	-1.16E+02	-2.62E+02	-2.58E+02	-2.54E+02
	vtlin(V)	-3.18E-01	-3.37E-01	-3.45E-01	-3.48E-01	-3.00E-01	-3.29E-01	-3.42E-01	-3.47E-01
	vtsat(V)	-2.85E-01	-2.31E-01	-2.42E-01	-2.45E-01	-2.67E-01	-2.27E-01	-2.43E-01	-2.49E-01
	body_effect(V)	-1.82E-01	-9.92E-02	-9.88E-02	-9.88E-02	-1.80E-01	-1.39E-01	-1.38E-01	-1.38E-01
	ioff(pA/um)	-9.21E+01	-1.80E+03	-1.42E+03	-1.41E+03	-2.02E+02	-2.36E+03	-1.48E+03	-1.20E+03
FF	ft(GHz)	19.202	71.182	63.338	54.455	NA	NA	NA	NA
	idsat(uA/um)	-1.52E+02	-3.53E+02	-3.74E+02	-3.65E+02	-1.52E+02	-3.54E+02	-3.59E+02	-3.58E+02
	vtlin(V)	-2.33E-01	-2.21E-01	-2.19E-01	-2.17E-01	-2.11E-01	-2.15E-01	-2.14E-01	-2.14E-01
	vtsat(V)	-2.00E-01	-1.01E-01	-1.02E-01	-1.01E-01	-1.77E-01	-1.10E-01	-1.11E-01	-1.12E-01
	body_effect(V)	-1.81E-01	-8.04E-02	-8.11E-02	-8.20E-02	-1.76E-01	-1.33E-01	-1.33E-01	-1.34E-01
	ioff(pA/um)	-1.27E+03	-5.55E+04	-5.69E+04	-6.22E+04	-2.82E+03	-4.64E+04	-4.38E+04	-4.21E+04

Table 2-5 RF & Logic Corner Table at 25 ℃

Temp=125C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	4	16	22	16	4	16
SS	ft(GHz)	13.254	42.451	38.961	34.595	NA	NA	NA	NA
	idsat(uA/um)	-1.00E+02	-2.12E+02	-2.17E+02	-2.12E+02	-9.50E+01	-2.14E+02	-2.04E+02	-1.98E+02
	vtlin(V)	-2.91E-01	-3.14E-01	-3.26E-01	-3.31E-01	-2.71E-01	-3.08E-01	-3.27E-01	-3.34E-01
	vtsat(V)	-2.51E-01	-2.06E-01	-2.21E-01	-2.26E-01	-2.32E-01	-2.03E-01	-2.23E-01	-2.32E-01
	body_effect(V)	-1.84E-01	-1.05E-01	-1.04E-01	-1.04E-01	-1.85E-01	-1.46E-01	-1.45E-01	-1.45E-01
	ioff(pA/um)	-1.47E+03	-1.20E+04	-9.04E+03	-8.39E+03	-2.55E+03	-1.43E+04	-9.23E+03	-7.59E+03
SNFP	ft(GHz)	16.693	59.944	55.09	49.16	NA	NA	NA	NA
	idsat(uA/um)	-1.28E+02	-2.95E+02	-3.11E+02	-3.05E+02	-1.26E+02	-3.02E+02	-2.95E+02	-2.91E+02
	vtlin(V)	-2.00E-01	-1.96E-01	-1.95E-01	-1.93E-01	-1.79E-01	-1.92E-01	-1.96E-01	-1.98E-01
	vtsat(V)	-1.60E-01	-7.48E-02	-7.77E-02	-7.71E-02	-1.40E-01	-7.88E-02	-8.50E-02	-8.79E-02
	body_effect(V)	-1.83E-01	-8.76E-02	-8.73E-02	-8.76E-02	-1.82E-01	-1.38E-01	-1.37E-01	-1.38E-01
	ioff(pA/um)	-1.12E+04	-1.55E+05	-1.48E+05	-1.52E+05	-1.86E+04	-1.49E+05	-1.31E+05	-1.23E+05
TT	ft(GHz)	15.496	53.854	49.854	44.618	NA	NA	NA	NA
	idsat(uA/um)	-1.19E+02	-2.66E+02	-2.79E+02	-2.74E+02	-1.15E+02	-2.73E+02	-2.66E+02	-2.61E+02
	vtlin(V)	-2.30E-01	-2.35E-01	-2.38E-01	-2.38E-01	-2.09E-01	-2.31E-01	-2.39E-01	-2.43E-01
	vtsat(V)	-1.90E-01	-1.19E-01	-1.26E-01	-1.27E-01	-1.70E-01	-1.19E-01	-1.30E-01	-1.35E-01
	body_effect(V)	-1.83E-01	-9.40E-02	-9.39E-02	-9.38E-02	-1.81E-01	-1.39E-01	-1.38E-01	-1.38E-01
	ioff(pA/um)	-5.79E+03	-6.65E+04	-5.95E+04	-5.93E+04	-9.71E+03	-7.10E+04	-5.66E+04	-5.10E+04
FNSP	ft(GHz)	14.323	48.195	44.556	39.857	NA	NA	NA	NA
	idsat(uA/um)	-1.09E+02	-2.38E+02	-2.47E+02	-2.43E+02	-1.05E+02	-2.44E+02	-2.35E+02	-2.30E+02
	vtlin(V)	-2.59E-01	-2.77E-01	-2.84E-01	-2.86E-01	-2.40E-01	-2.70E-01	-2.83E-01	-2.88E-01
	vtsat(V)	-2.20E-01	-1.65E-01	-1.75E-01	-1.78E-01	-2.01E-01	-1.60E-01	-1.76E-01	-1.83E-01
	body_effect(V)	-1.84E-01	-1.00E-01	-9.97E-02	-9.95E-02	-1.81E-01	-1.41E-01	-1.40E-01	-1.40E-01
	ioff(pA/um)	-2.94E+03	-2.71E+04	-2.24E+04	-2.16E+04	-4.98E+03	-3.23E+04	-2.32E+04	-2.00E+04
FF	ft(GHz)	17.559	65.818	60.12	53.898	NA	NA	NA	NA
	idsat(uA/um)	-1.39E+02	-3.29E+02	-3.49E+02	-3.44E+02	-1.37E+02	-3.31E+02	-3.26E+02	-3.22E+02
	vtlin(V)	-1.73E-01	-1.60E-01	-1.57E-01	-1.54E-01	-1.48E-01	-1.53E-01	-1.52E-01	-1.52E-01
	vtsat(V)	-1.34E-01	-3.46E-02	-3.50E-02	-3.40E-02	-1.10E-01	-4.09E-02	-4.20E-02	-4.29E-02
	body_effect(V)	-1.83E-01	-8.14E-02	-8.18E-02	-8.23E-02	-1.78E-01	-1.36E-01	-1.35E-01	-1.36E-01
	ioff(pA/um)	-1.97E+04	-3.23E+05	-3.23E+05	-3.31E+05	-3.47E+04	-2.93E+05	-2.85E+05	-2.80E+05

Table 2-6 RF & Logic Corner Table at 125 ℃

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	4	16	22	16	4	16
SS	ft(GHz)	16.885	51.429	45.609	38.612	NA	NA	NA	NA
	idsat(uA/um)	-1.23E+02	-2.50E+02	-2.54E+02	-2.44E+02	-1.19E+02	-2.54E+02	-2.52E+02	-2.48E+02
	vtlin(V)	-4.02E-01	-4.27E-01	-4.43E-01	-4.50E-01	-3.83E-01	-4.17E-01	-4.35E-01	-4.43E-01
	vtsat(V)	-3.71E-01	-3.27E-01	-3.45E-01	-3.51E-01	-3.52E-01	-3.21E-01	-3.42E-01	-3.51E-01
	body_effect(V)	-1.81E-01	-1.04E-01	-1.03E-01	-1.03E-01	-1.82E-01	-1.46E-01	-1.45E-01	-1.45E-01
	ioff(pA/um)	-5.66E-01	-1.07E+01	-7.82E+00	-7.91E+00	-1.13E+00	-1.71E+01	-8.48E+00	-6.73E+00
SNFP	ft(GHz)	21.179	72.158	63.914	54.028	NA	NA	NA	NA
	idsat(uA/um)	-1.61E+02	-3.50E+02	-3.66E+02	-3.55E+02	-1.60E+02	-3.49E+02	-3.62E+02	-3.65E+02
	vtlin(V)	-3.12E-01	-3.10E-01	-3.13E-01	-3.13E-01	-2.94E-01	-3.05E-01	-3.09E-01	-3.11E-01
	vtsat(V)	-2.81E-01	-1.97E-01	-2.02E-01	-2.02E-01	-2.63E-01	-2.02E-01	-2.09E-01	-2.12E-01
	body_effect(V)	-1.80E-01	-8.69E-02	-8.73E-02	-8.79E-02	-1.79E-01	-1.37E-01	-1.36E-01	-1.37E-01
	ioff(pA/um)	-1.12E+01	-1.29E+03	-1.27E+03	-1.53E+03	-3.06E+01	-1.14E+03	-8.66E+02	-7.54E+02
TT	ft(GHz)	19.638	64.77	57.743	49.019	NA	NA	NA	NA
	idsat(uA/um)	-1.48E+02	-3.15E+02	-3.28E+02	-3.17E+02	-1.46E+02	-3.17E+02	-3.26E+02	-3.26E+02
	vtlin(V)	-3.41E-01	-3.49E-01	-3.56E-01	-3.59E-01	-3.24E-01	-3.42E-01	-3.51E-01	-3.55E-01
	vtsat(V)	-3.10E-01	-2.41E-01	-2.50E-01	-2.52E-01	-2.92E-01	-2.41E-01	-2.52E-01	-2.57E-01
	body_effect(V)	-1.81E-01	-9.32E-02	-9.31E-02	-9.35E-02	-1.79E-01	-1.38E-01	-1.37E-01	-1.38E-01
	ioff(pA/um)	-3.49E+00	-2.56E+02	-2.20E+02	-2.46E+02	-9.44E+00	-2.93E+02	-1.87E+02	-1.51E+02
FNSP	ft(GHz)	18.129	58.014	51.644	43.843	NA	NA	NA	NA
	idsat(uA/um)	-1.35E+02	-2.81E+02	-2.89E+02	-2.80E+02	-1.32E+02	-2.85E+02	-2.89E+02	-2.87E+02
	vtlin(V)	-3.70E-01	-3.90E-01	-4.02E-01	-4.06E-01	-3.53E-01	-3.80E-01	-3.93E-01	-3.99E-01
	vtsat(V)	-3.40E-01	-2.86E-01	-2.99E-01	-3.04E-01	-3.22E-01	-2.81E-01	-2.97E-01	-3.04E-01
	body_effect(V)	-1.81E-01	-9.93E-02	-9.89E-02	-9.91E-02	-1.79E-01	-1.40E-01	-1.39E-01	-1.39E-01
	ioff(pA/um)	-1.25E+00	-4.71E+01	-3.56E+01	-3.71E+01	-3.03E+00	-7.05E+01	-3.82E+01	-2.93E+01
FF	ft(GHz)	22.076	78.945	69.616	58.806	NA	NA	NA	NA
	idsat(uA/um)	-1.74E+02	-3.87E+02	-4.09E+02	-3.95E+02	-1.75E+02	-3.81E+02	-3.99E+02	-4.03E+02
	vtlin(V)	-2.85E-01	-2.76E-01	-2.76E-01	-2.75E-01	-2.65E-01	-2.67E-01	-2.67E-01	-2.67E-01
	vtsat(V)	-2.54E-01	-1.57E-01	-1.59E-01	-1.59E-01	-2.33E-01	-1.65E-01	-1.67E-01	-1.69E-01
	body_effect(V)	-1.80E-01	-8.10E-02	-8.18E-02	-8.27E-02	-1.76E-01	-1.35E-01	-1.34E-01	-1.34E-01
	ioff(pA/um)	-3.12E+01	-5.15E+03	-5.50E+03	-6.58E+03	-9.51E+01	-3.90E+03	-3.56E+03	-3.35E+03

Table 2-7 RF & Logic Corner Table at -50 ℃

2.2.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted Idsat is smaller than the real Idsat for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
PFET601	0.12	1.6	16	6	1	-0.291	-0.301	-0.182	-0.194	-283.26	-285.58	-6347.95	-5778.8	5.133	43.708
PFET602	0.12	2.4	16	4	1	-0.293	-0.3	-0.184	-0.192	-279.57	-290.88	-6034.87	-5504	47.58	50.209
PFET603	0.12	4.8	16	2	1	-0.293	-0.298	-0.182	-0.189	-276.37	-282.88	-6707.18	-5792.2	51.419	54.119
PFET604	0.12	9.6	16	1	1	-0.293	-0.296	-0.18	-0.185	-266.52	-270.21	-6994.5	-6264.2	53.906	53.737
PFET605	0.15	2.4	16	4	1	-0.309	-0.317	-0.249	-0.249	-204.89	-206.04	-478.07	-718.12	33.454	33.412
PFET606	0.18	2.4	16	4	1	-0.298	-0.311	-0.259	-0.262	-168.21	-166.48	-244.26	-341.28	24.386	24.456
PFET607	0.24	2.4	16	4	1	-0.281	-0.294	-0.249	-0.26	-127.23	-125.12	-216.54	-214.72	15.014	15.059
PFET608	0.36	2.4	16	4	1	-0.271	-0.274	-0.243	-0.249	-85.73	-85.634	-170.95	-168.78	7.551	7.514
PFET609	0.12	2.4	2	18	1	-0.296	-0.3	-0.189	-0.192	-295.7	-294.58	-5417.44	-5504.5	46.418	50.873
PFET611	0.12	2.4	5	13	1	-0.293	-0.3	-0.183	-0.192	-287.54	-292.11	-6255.98	-5504.1	47.667	50.026
PFET612	0.12	2.4	8	8	1	-0.294	-0.3	-0.185	-0.192	-282.55	-291.71	-6104.17	-5504	48.033	50.124
PFET613	0.12	2.4	11	6	1	-0.294	-0.3	-0.184	-0.192	-281.21	-290.54	-6152.08	-5504	48.274	49.637
PFET619	0.12	4.8	4	8	1	-0.294	-0.298	-0.185	-0.189	-287.31	-285.61	-6000.14	-5792.3	52.708	54.1
PFET620	0.12	4.8	11	2	1	-0.292	-0.298	-0.18	-0.189	-282.34	-284.48	-7069.65	-5792.2	54.346	54.422
PFET621	0.12	4.8	22	1	1	-0.292	-0.298	-0.18	-0.189	-280.24	-282.1	-6801.56	-5792.2	53.636	53.807
PFET622	0.15	9.6	4	4	1	-0.31	-0.313	-0.251	-0.244	-208	-201.81	-457.13	-772.43	37.407	36.829
PFET623	0.18	9.6	11	1	1	-0.299	-0.306	-0.259	-0.257	-167.15	-164.85	-250.76	-366.06	27.203	26.892
PFET624	0.24	9.6	22	1	1	-0.284	-0.289	-0.254	-0.255	-124.66	-125.77	-191.41	-233.85	16.469	16.351
PFET625	0.12	2.4	4	1	1	-0.299	-0.3	-0.19	-0.192	-294.54	-295.44	-5068.52	-5504.2	53.058	51.491

Table 2-8 Summary Table for RF PMOS

Measure: Measurement result
Model: Simulation result

2.2.3 Simulation results for WF, LF trends

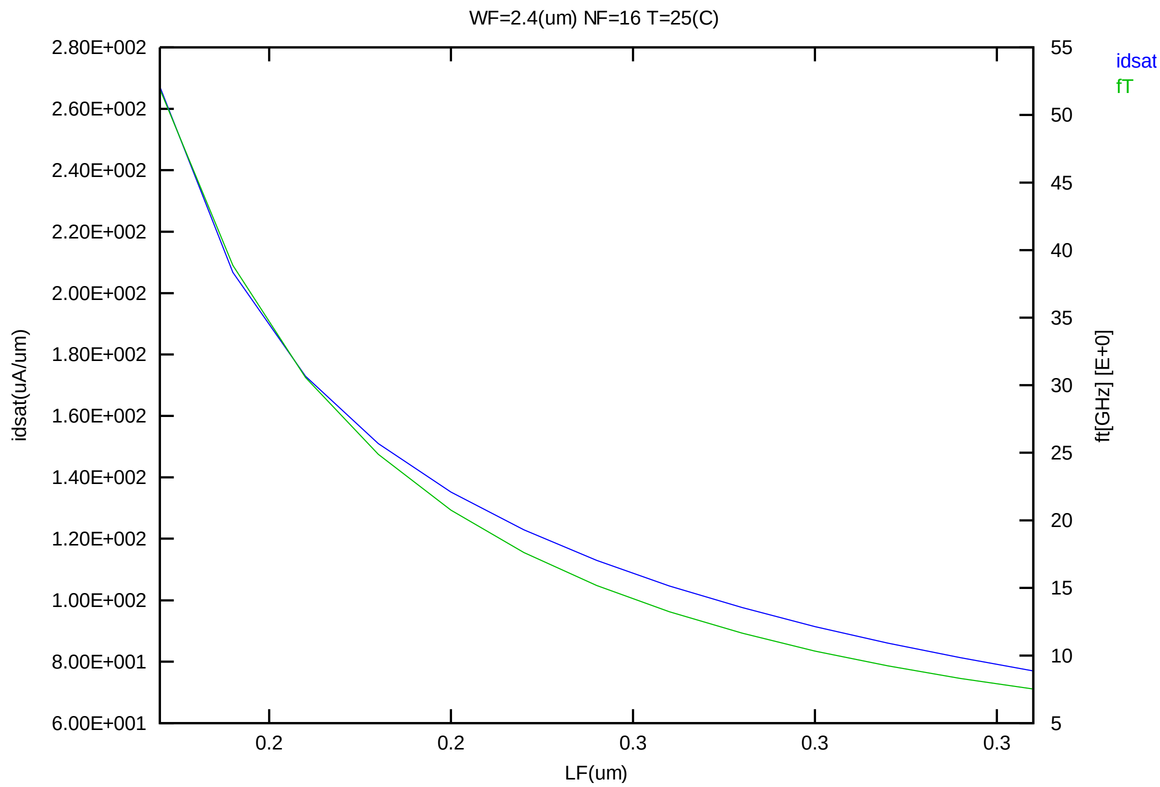


Figure 2-14 fT and Idsat vs. channel length

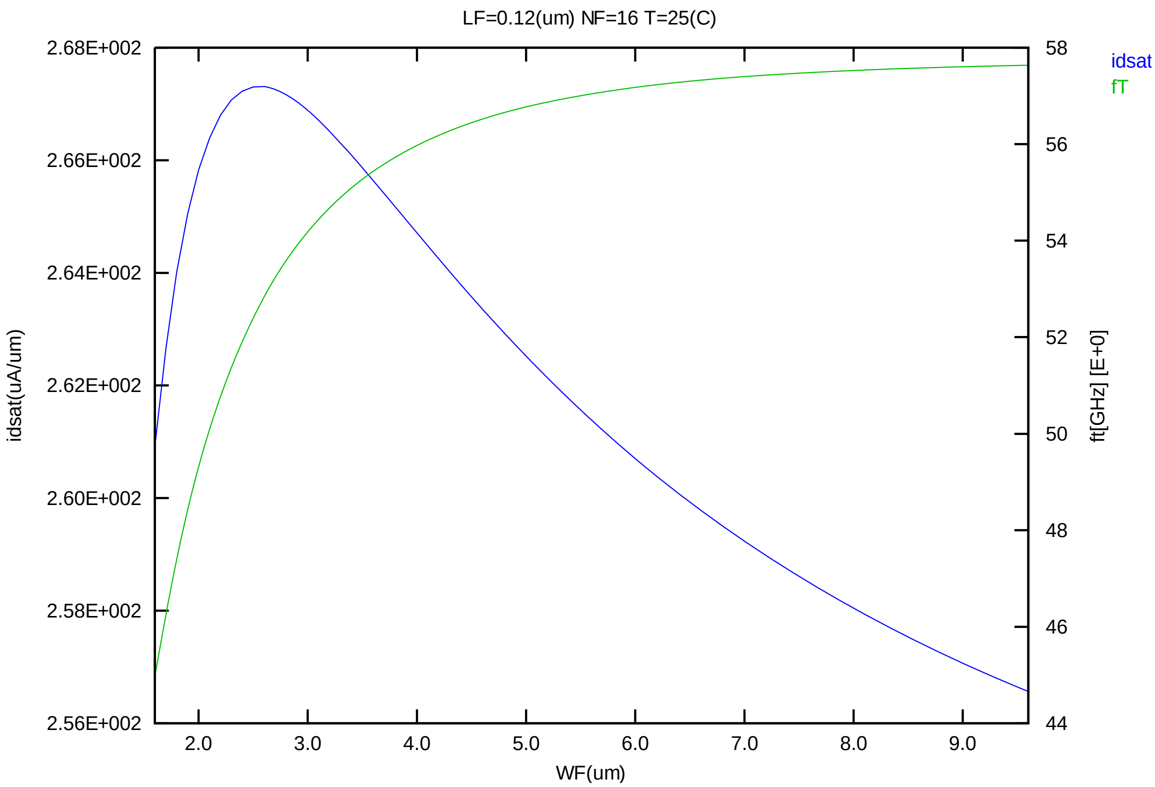
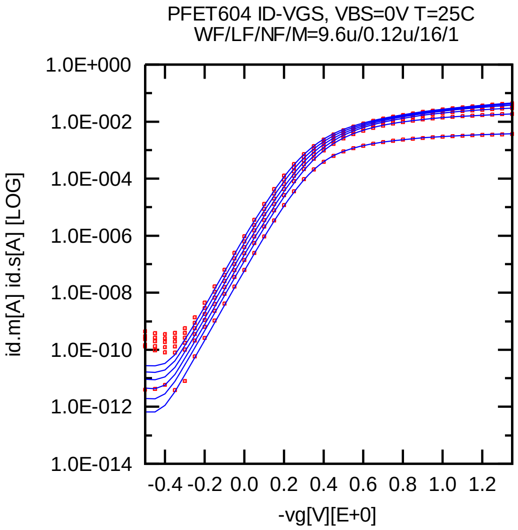
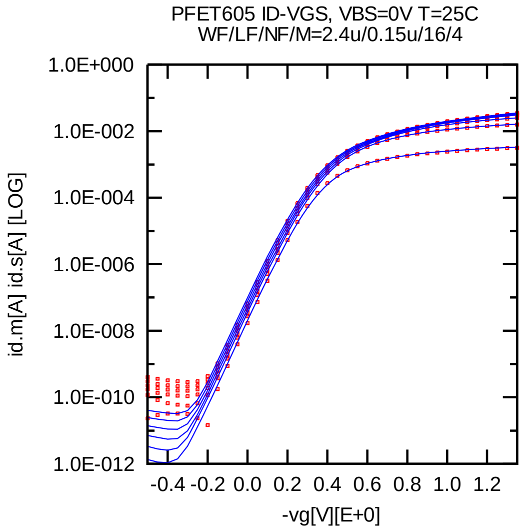
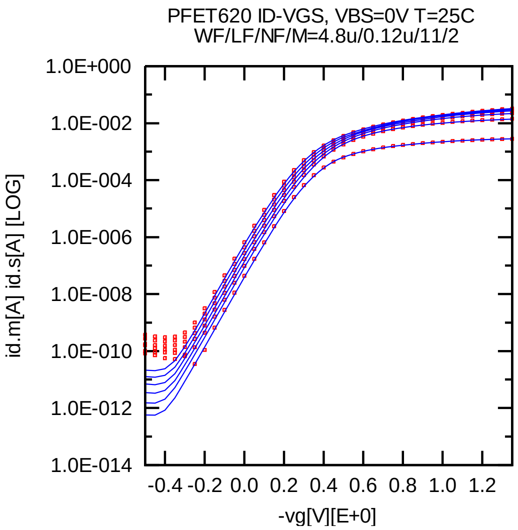


Figure 2-15 fT and Idsat vs. channel width

2.2.4 DC & RF Fitting curves for individual devices







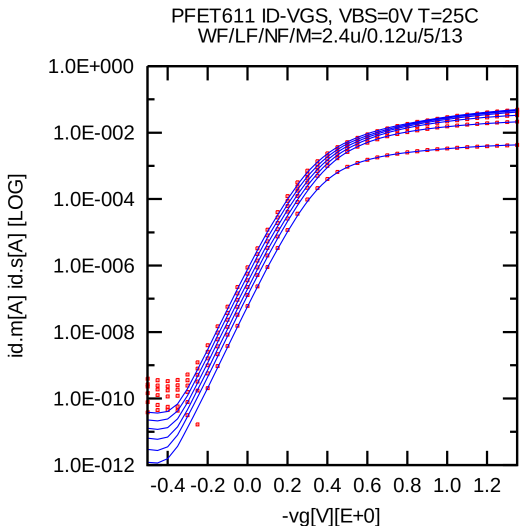


Figure 2-16 Id-Vg (25℃)

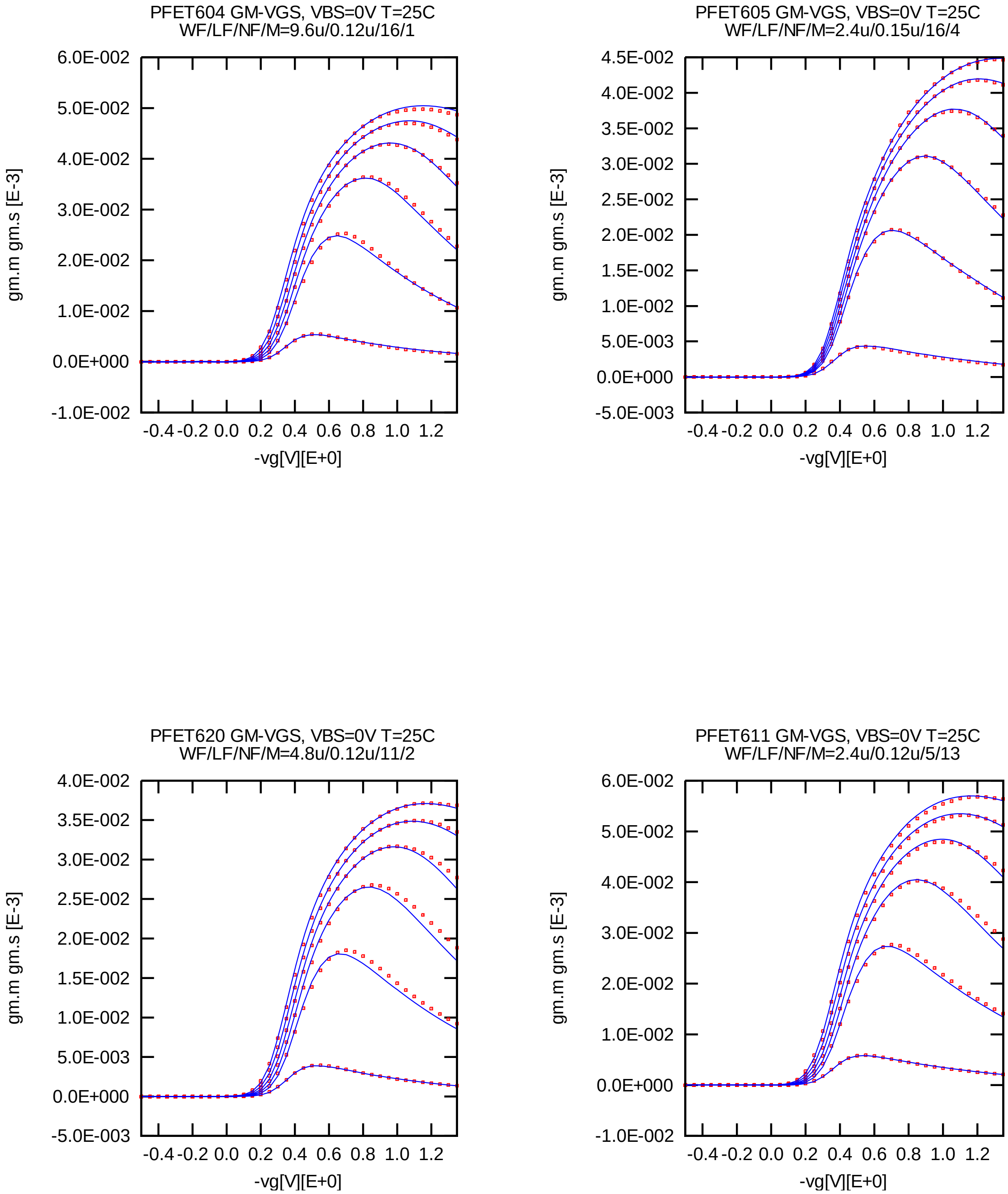


Figure 2-17 Gm-Vg (25℃)

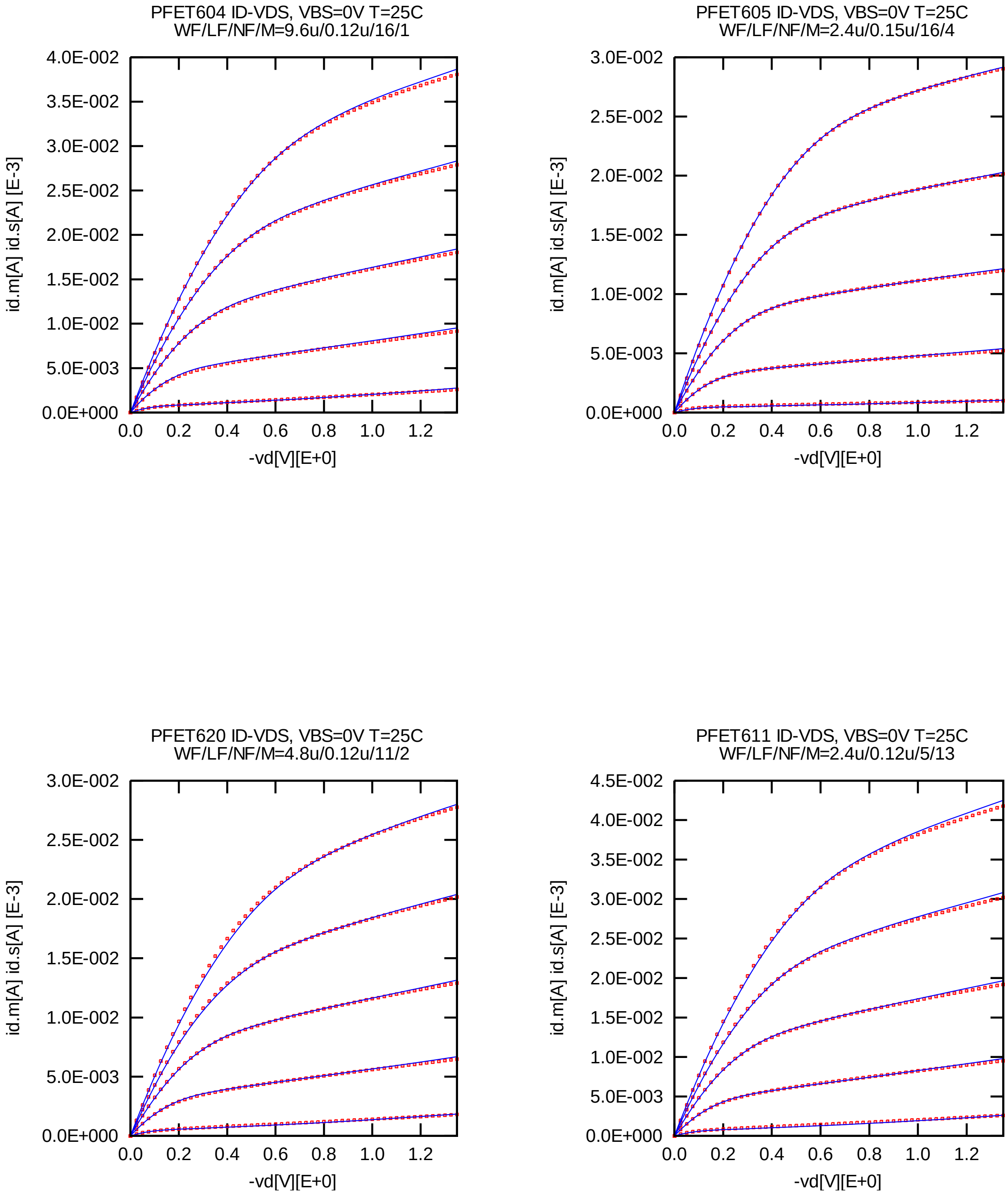
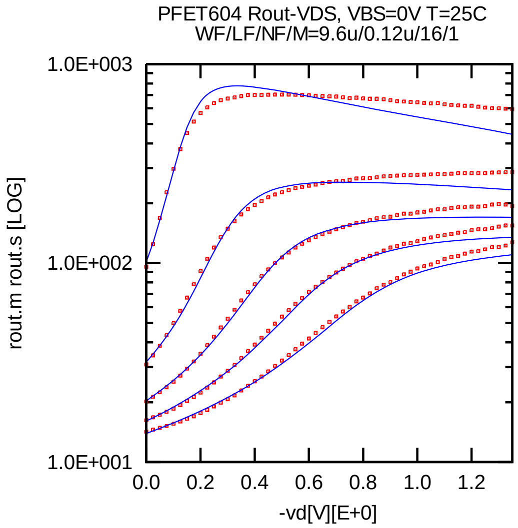
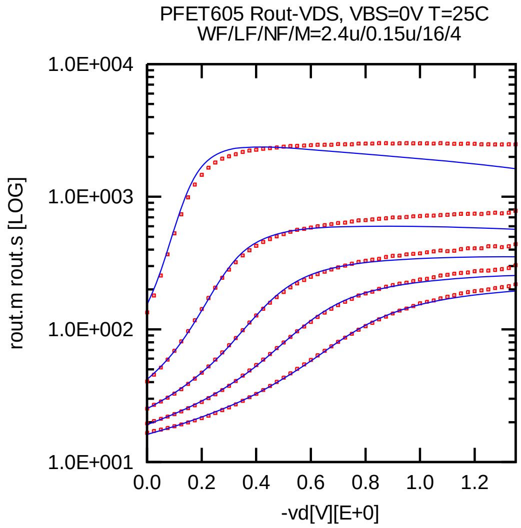
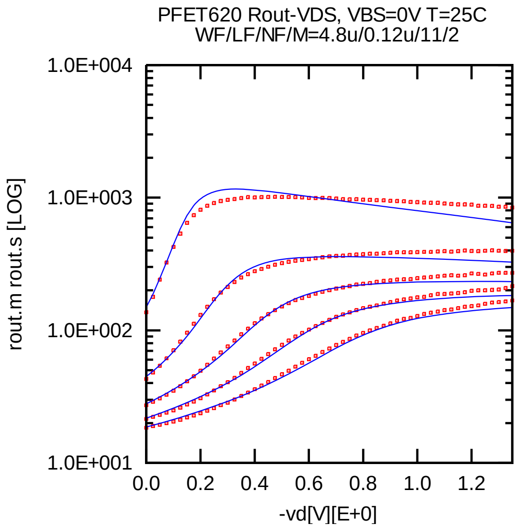


Figure 2-18 Id-Vd (25℃)







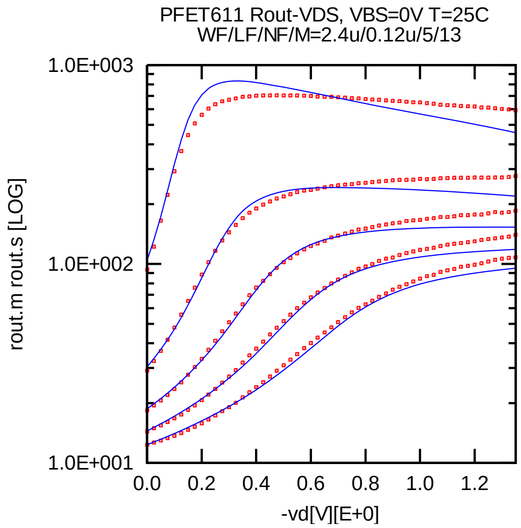


Figure 2-19 Rout-Vd (25℃)

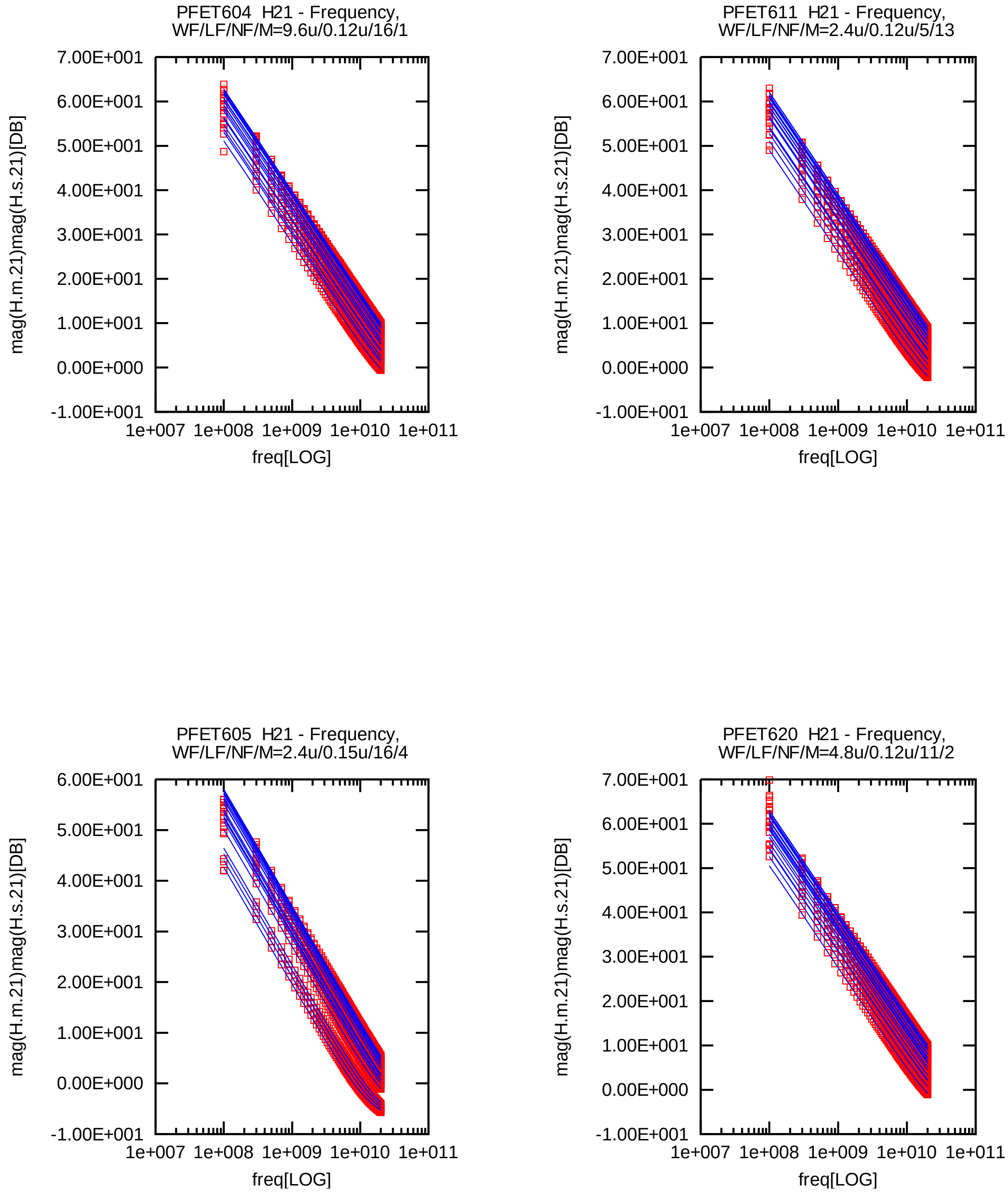


Figure 2-20 Mag(H21)-Freq (25℃)

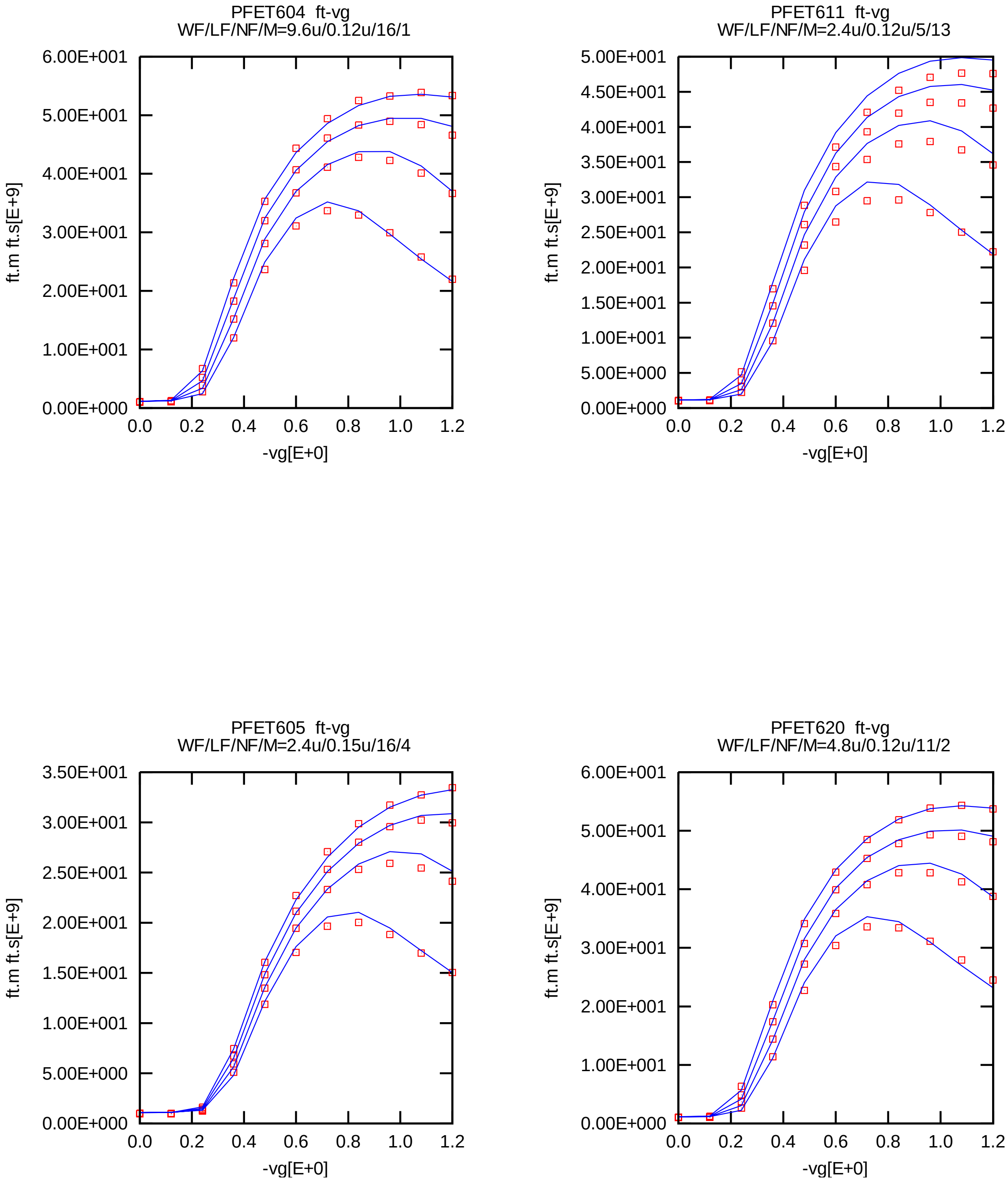


Figure 2-21 fT-Vg (25℃)

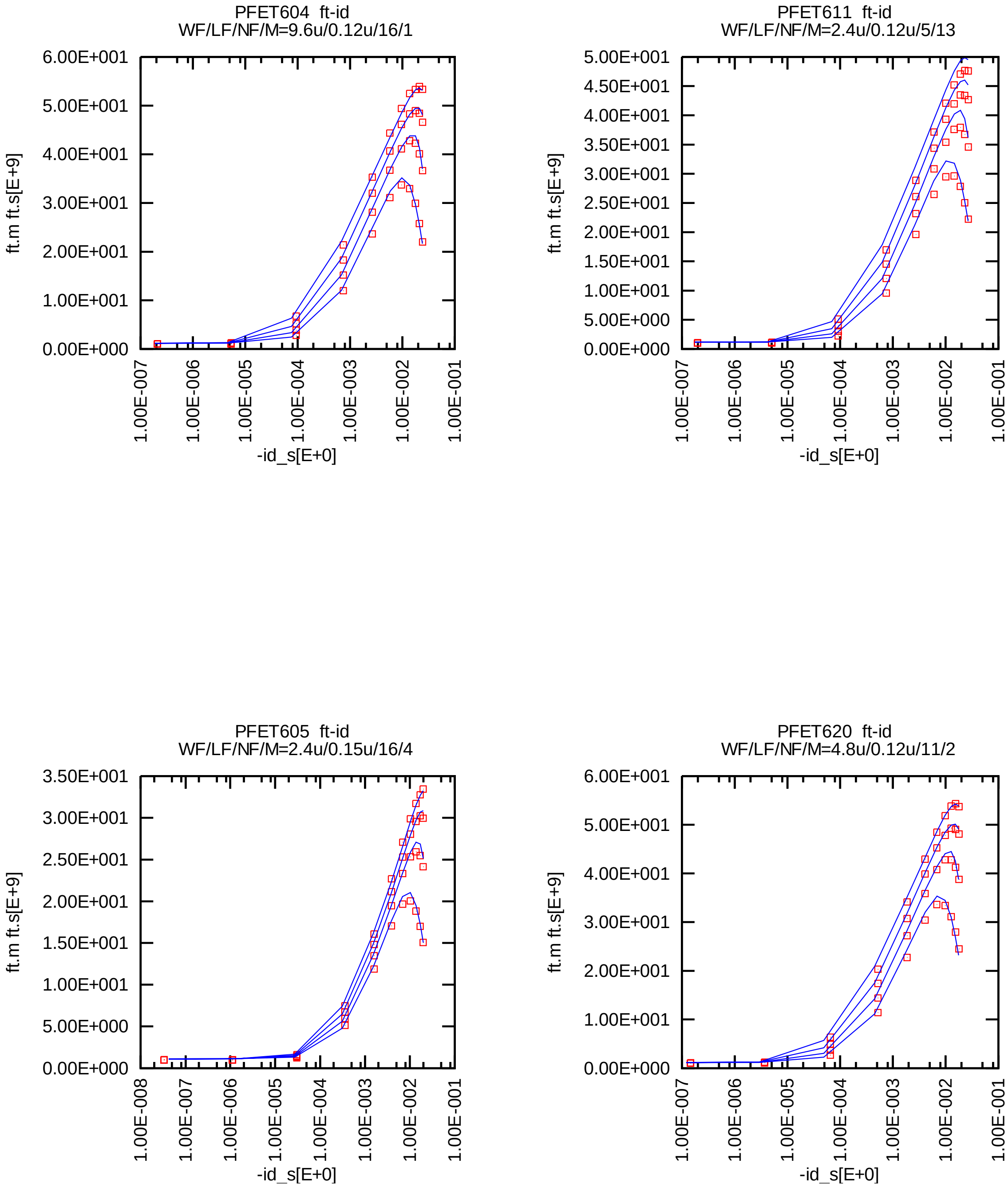


Figure 2-22 fT-Id (25℃)

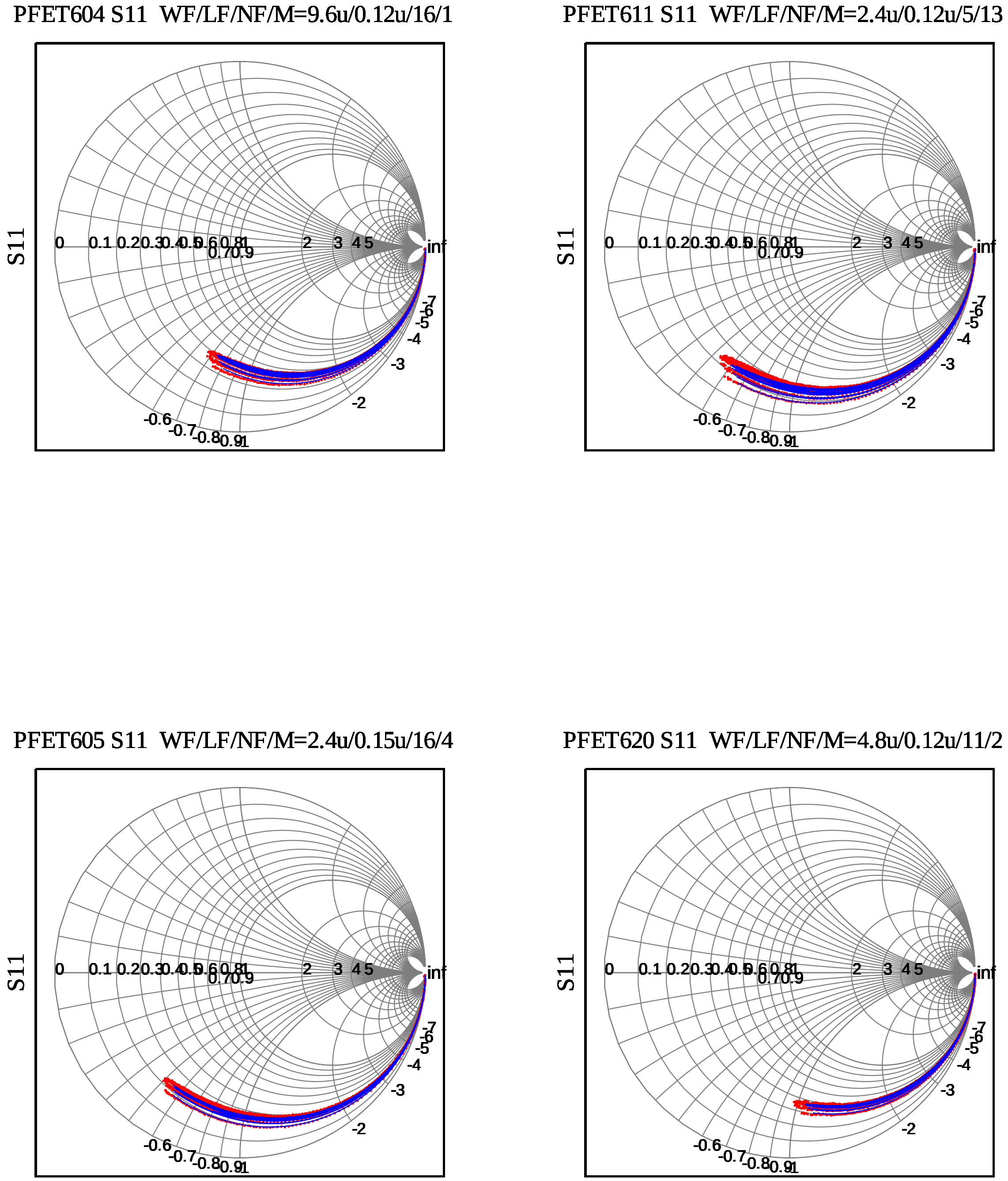


Figure 2-23 S11 (25℃)

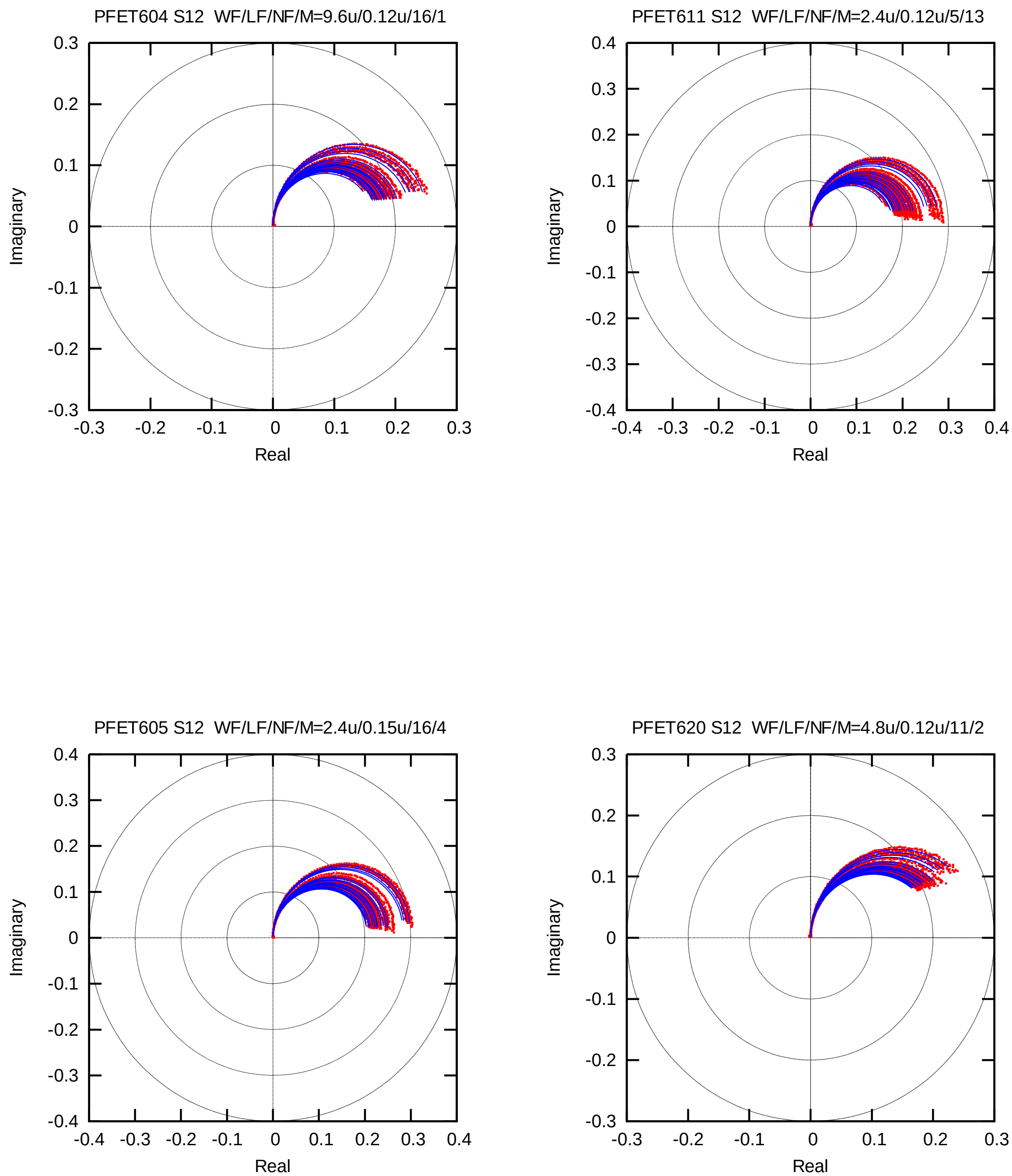


Figure 2-24 S12 (25℃)

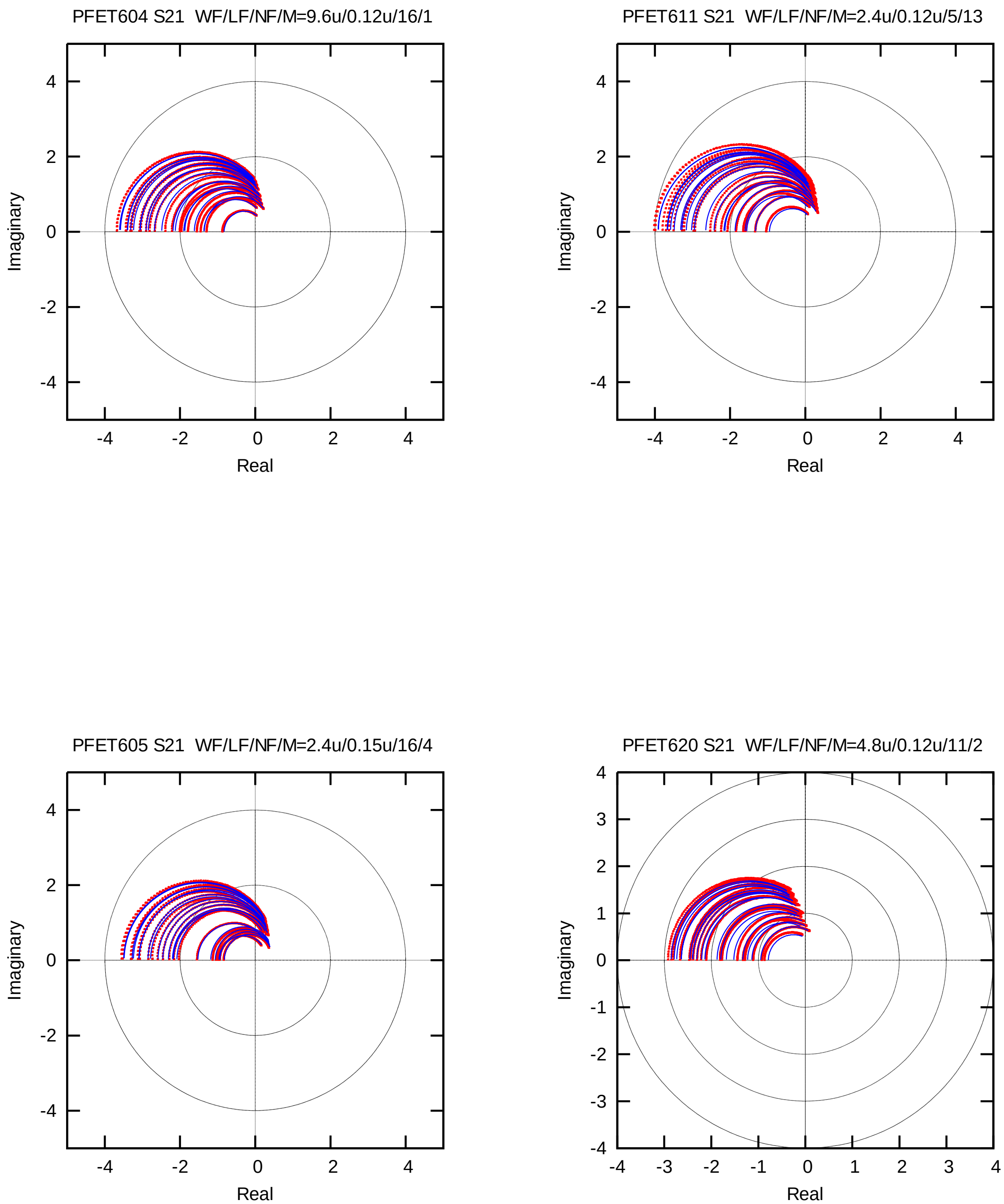
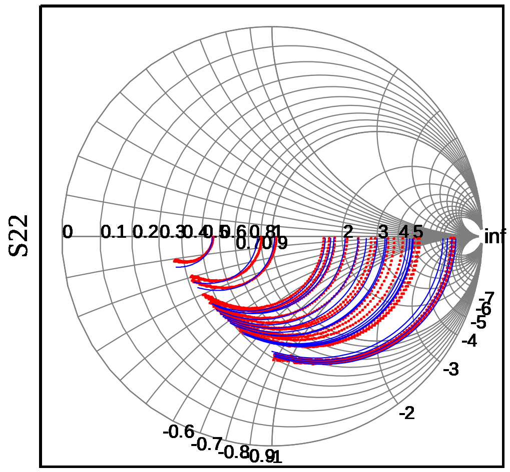
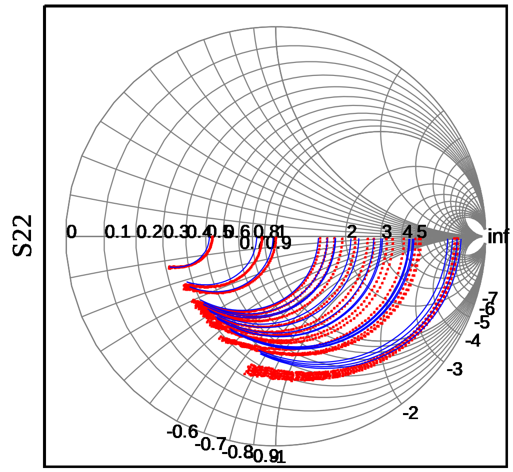


Figure 2-25 S21 (25℃)

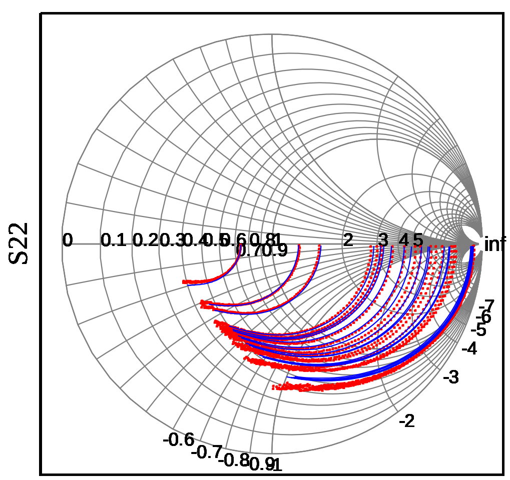
PFET604 S22 WF/LF/NF/M=9.6u/0.12u/16/1



PFET611 S22 WF/LF/NF/M=2.4u/0.12u/5/13



PFET605 S22 WF/LF/NF/M=2.4u/0.15u/16/4



PFET620 S22 WF/LF/NF/M=4.8u/0.12u/11/2

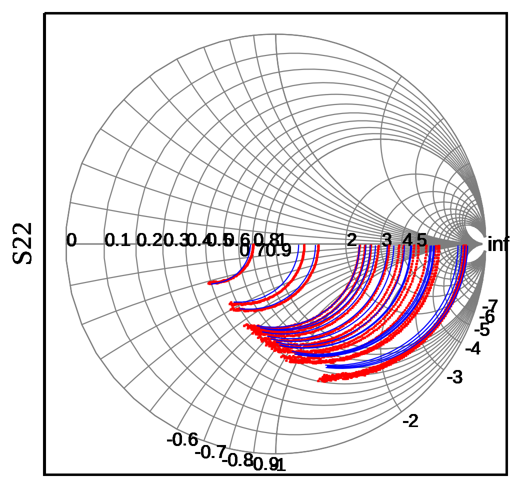


Figure 2-26 S22 (25℃)

3 High Frequency Noise Model Enhancement

3.1 General Information

This section describes the MOSFET RF models with High Frequency (HF) noise enhancement. At the current release, all four simulators (**Spectre, Hspice and Eldo)** models support HF noise model enhancement.

Except noise behavior, the RF models with and without HF noise enhancement have exact same electrical behaviors.

However, the simulation with the HF noise enhancement generally needs more simulation time than the simulation without HF noise enhancement since Verilog-A code is used in the HF noise enhancement. It is recommended to use the model without HF noise enhancement for simulations other than noise analysis, and use the RF model with HF noise enhancement for noise analysis to get more accurate noise results.

3.2 Model Usage

3.2.1 Voltage Scaling Range

The following voltage scaling range has been used to extract and verify high-frequency noise parameters:
Vg: 0.5 ~ 1.2V
Vd: 0.4 ~ 1.2V
Vs: 0V
Vb: 0V

If the device is biased outside the above voltage range, the HF noise enhancement may not accurately predict the noise performance.

3.2.2 Global and Local Flag Parameters for HF Noise Simulation

Two levels of flag parameters are designed to switch on or off the HF noise enhancement:

hfnoise (global flag): 1 is on and 0 is off

"*hfnoise*" need to be pre-defined. If using netlist, add the following line in the netlist:
parameters hfnoise=1 (Spectre)
.param hfnoise=1 (Hspice and Eldo)
If using Cadence Analog Environment, "hfnoise" need to be defined as a Design Variable before simulation.

hfn_flag (local flag): 1 is on and 0 is off

"*hfn_flag*" is a model input parameter. If it is not given in the netlist, its default value will equal to the global parameter "*hfnoise*"

The following examples demonstrate how to use these two flag parameters:

Example 1: HF noise enhancement switched off for transistors x1, x2, x3
Parameters hfnoise=0
x1 (d1 g1 s1 b1) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8
x2 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8
x3 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8

Example 2: HF noise enhancement switched on for transistors x1, x2, x3
Parameters hfnoise=1
x1 (d1 g1 s1 b1) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8
x2 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8
x3 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8

Example 3: HF noise enhancement switched off for transistors x2, x3 but switched on for x1
Parameters hfnoise=0
x1 (d1 g1 s1 b1) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8 hfn_flag=1
x2 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8
x3 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8 hfn_flag=0

Example 4: HF noise enhancement switched on for transistors x1, x2, but switched off for x3
Parameters hfnoise=1
x1 (d1 g1 s1 b1) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8 hfn_flag=1
x2 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8
x3 (d2 g2 s2 b2) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8 hfn_flag=0

3.2.3 Multiple Factor "mf" for Hspice and Eldo

Due to the fact that Verilog-A codes in Hspice and Eldo don't support multiple factor which is inherited from the netlist, an additional model input parameter "*mf*" has to be defined in order to accurately simulate the High Frequency noise for multiple devices connected in parallel. *mf* will not affect the simulation results other than thermal noise behavior. The value of *mf* should be always be equal to the value of instant multiple factor parameter *m*. Please refer to the following example:

If 4 devices are connected in parallel, the netlist for Hspice and Eldo should be:

x1 (d g s b) n_12_hsrf lf=0.12u wf=3.6u nf=4 m=8 mf=8

3.3 HF Noise Modeling Result

3.3.1 NMOS Noise Parameters – NF_{min} , R_n & G_{opt}

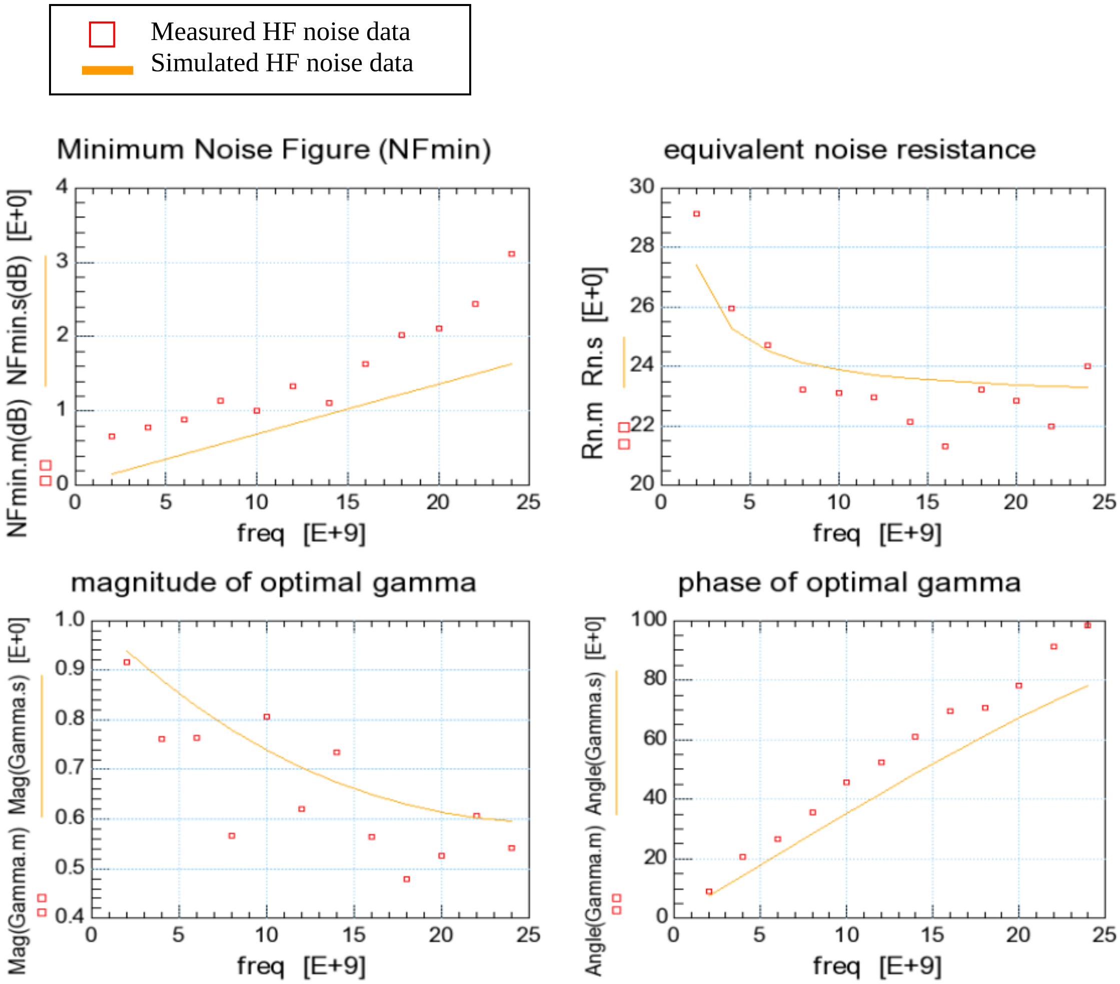


Figure 3-1 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.12um, WF=3.6um, NF=4, m=8 @ Vd=0.4, Vg=0.6V

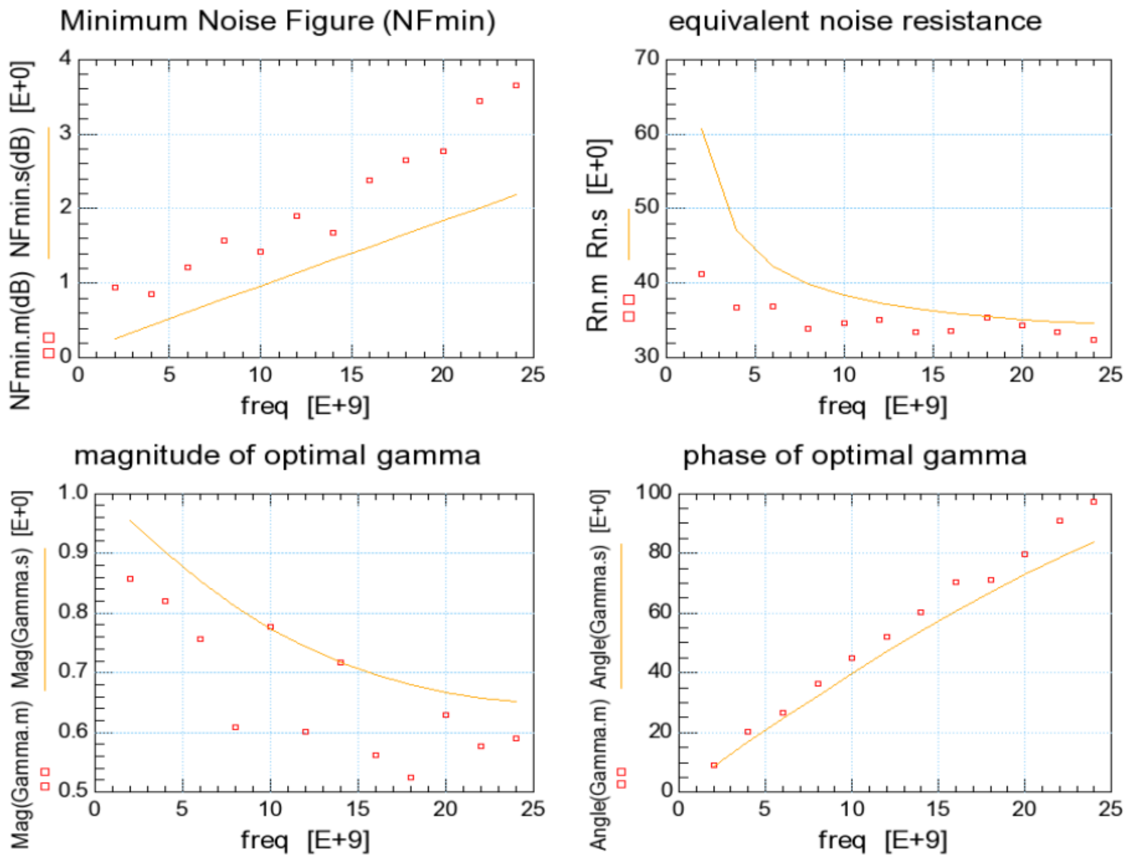


Figure 3-2 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.12um, WF=3.6um, NF=4, m=8 @ Vd=1.2, Vg=1.2V

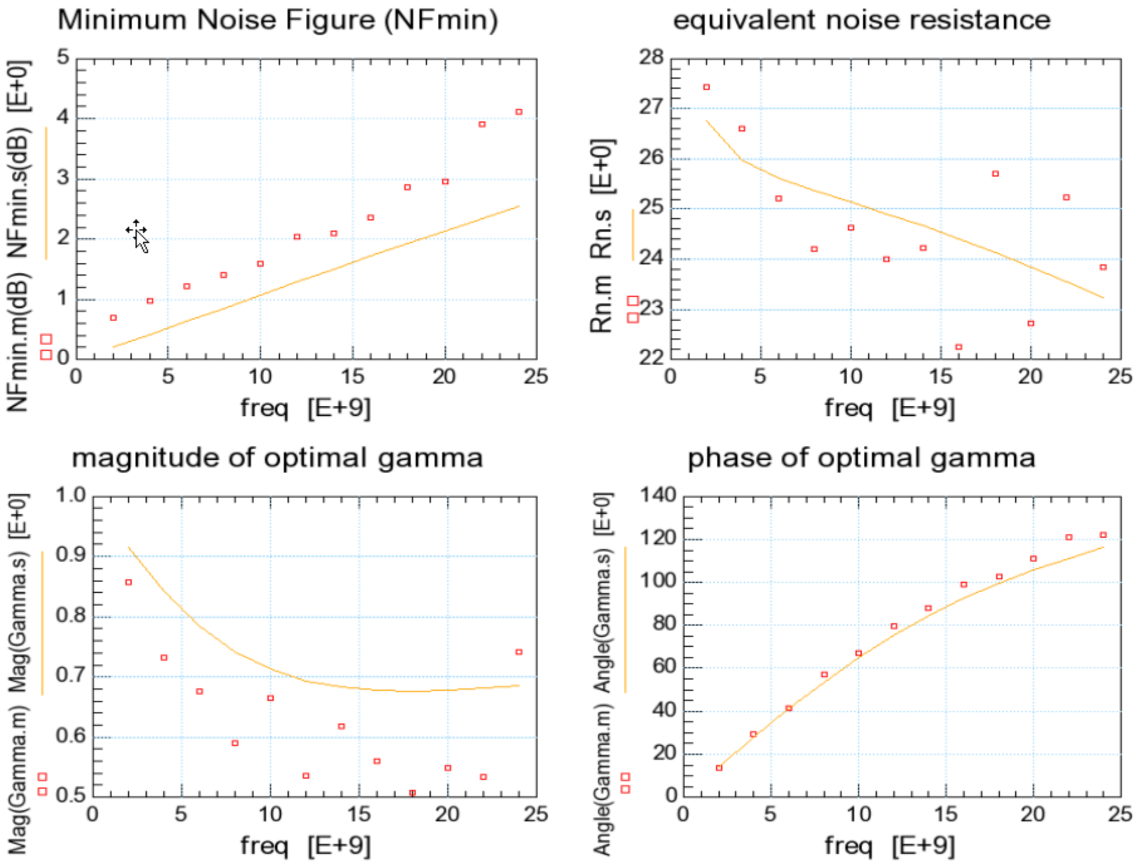


Figure 3-3 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.24um, WF=3.6um, NF=16, m=2 @ Vd=0.4, Vg=0.6V

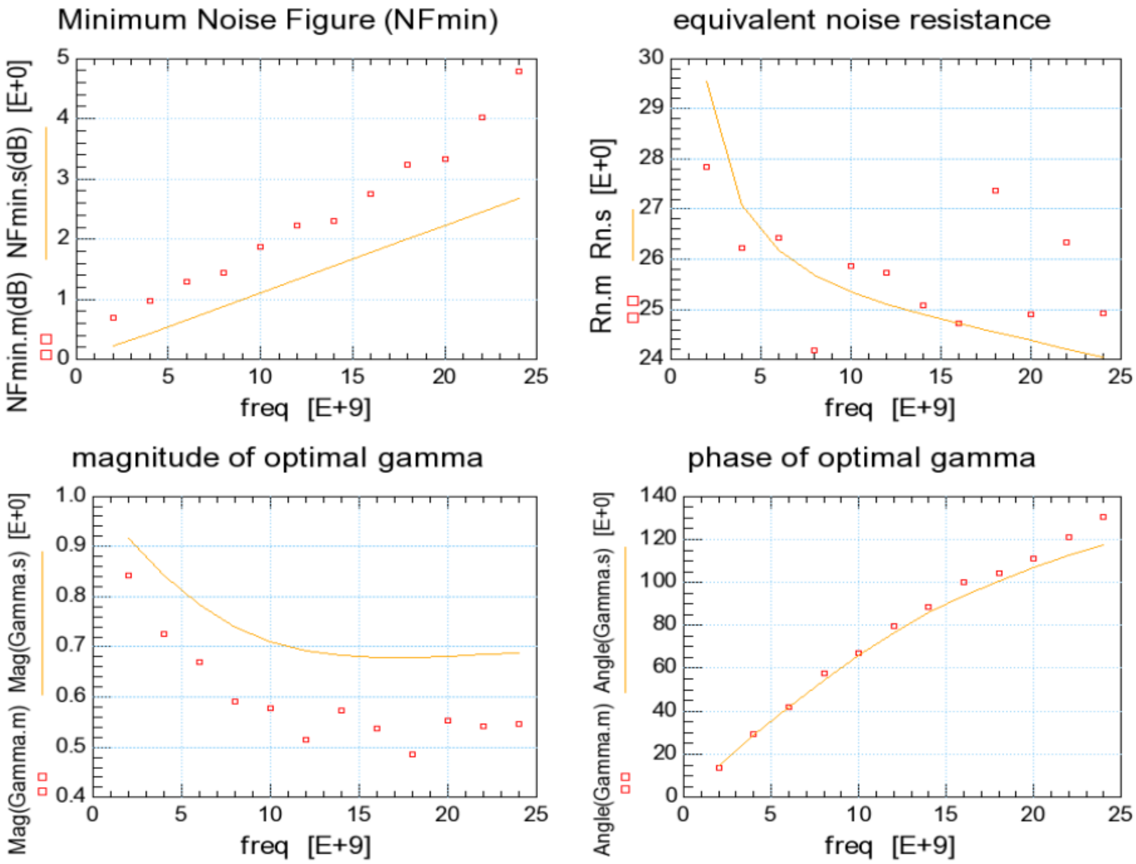


Figure 3-4 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.24um, WF=3.6um, NF=16, m=2 @ Vd=1.2, Vg=1V

3.3.2 PMOS Noise Parameters – NF_{min} , R_n & G_{opt}

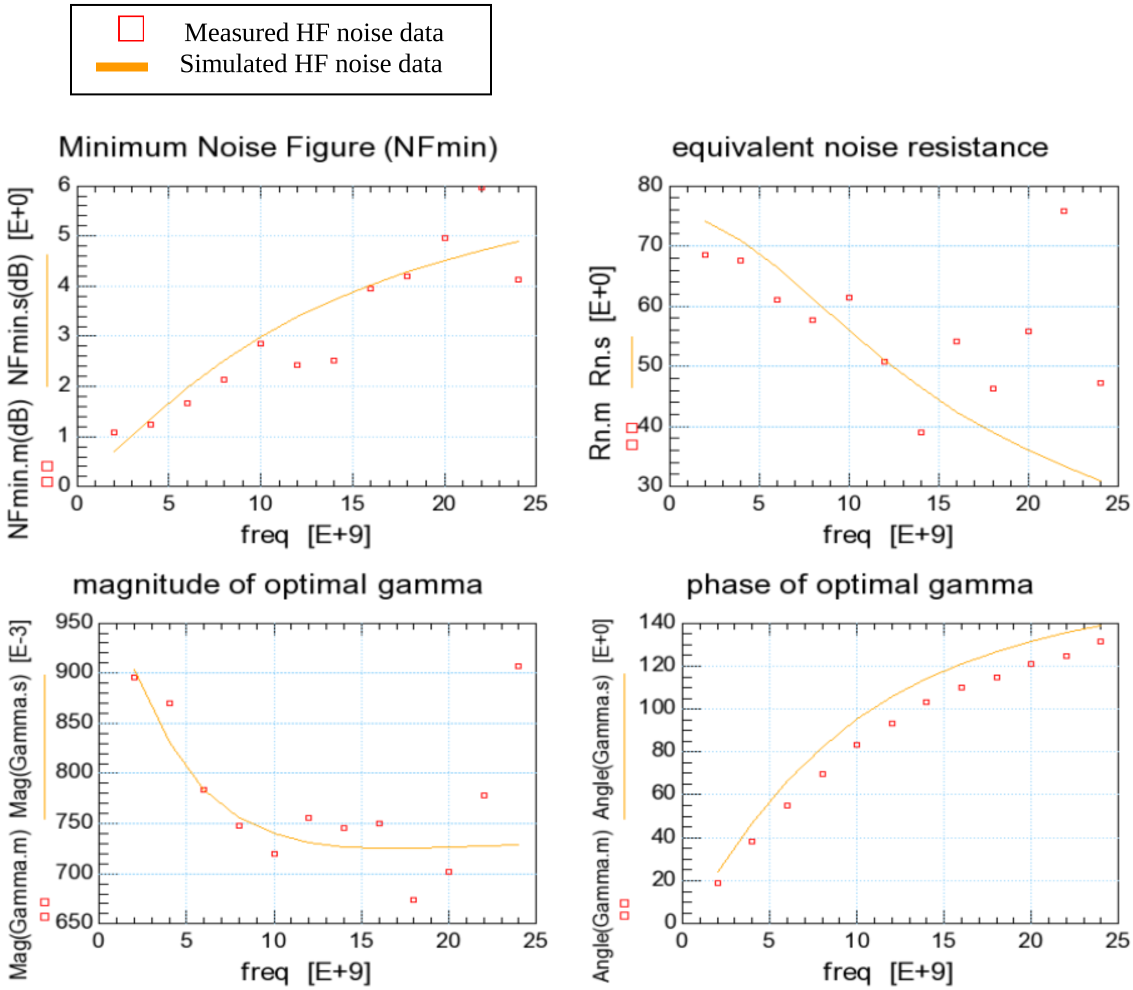


Figure 3-5 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.24um, WF=2.4um, NF=16, m=4 @ Vd=-0.4, Vg=-0.6V

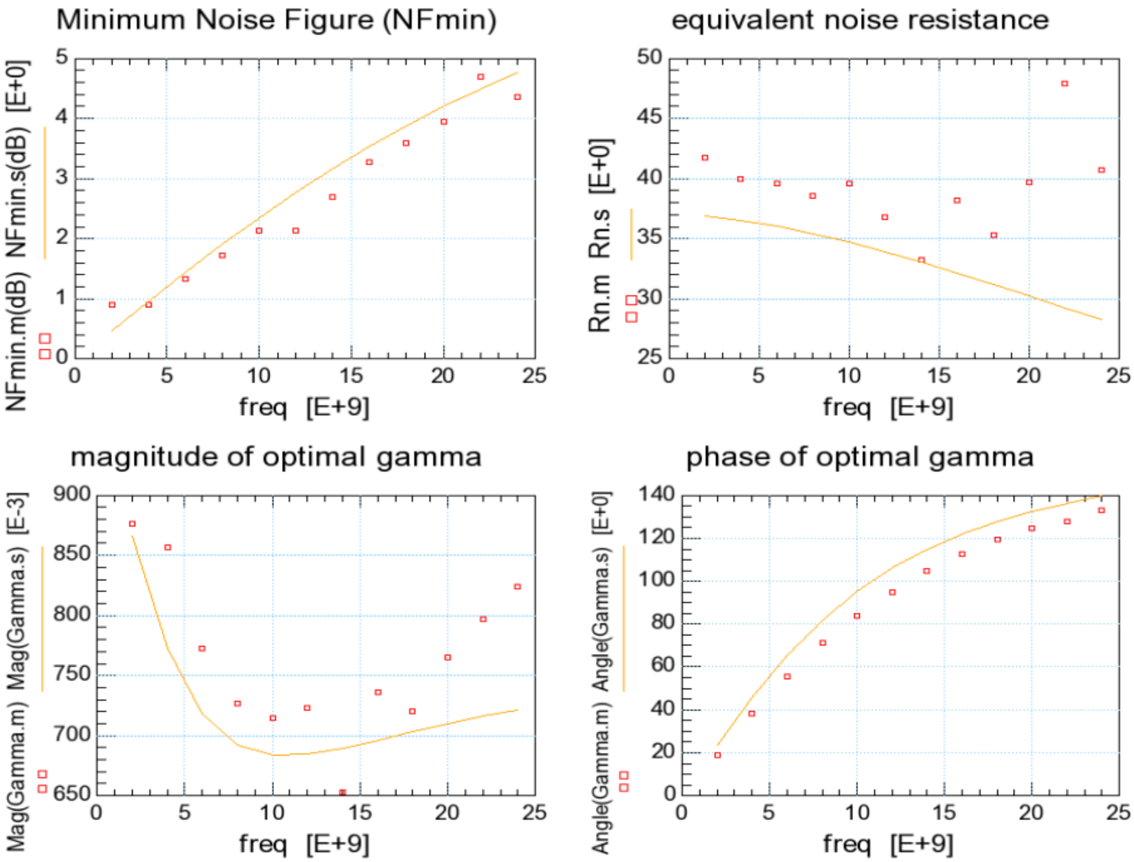


Figure 3-6 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.24um, WF=2.4um, NF=16, m=4 @ Vd=-1.2, Vg=-1.2V

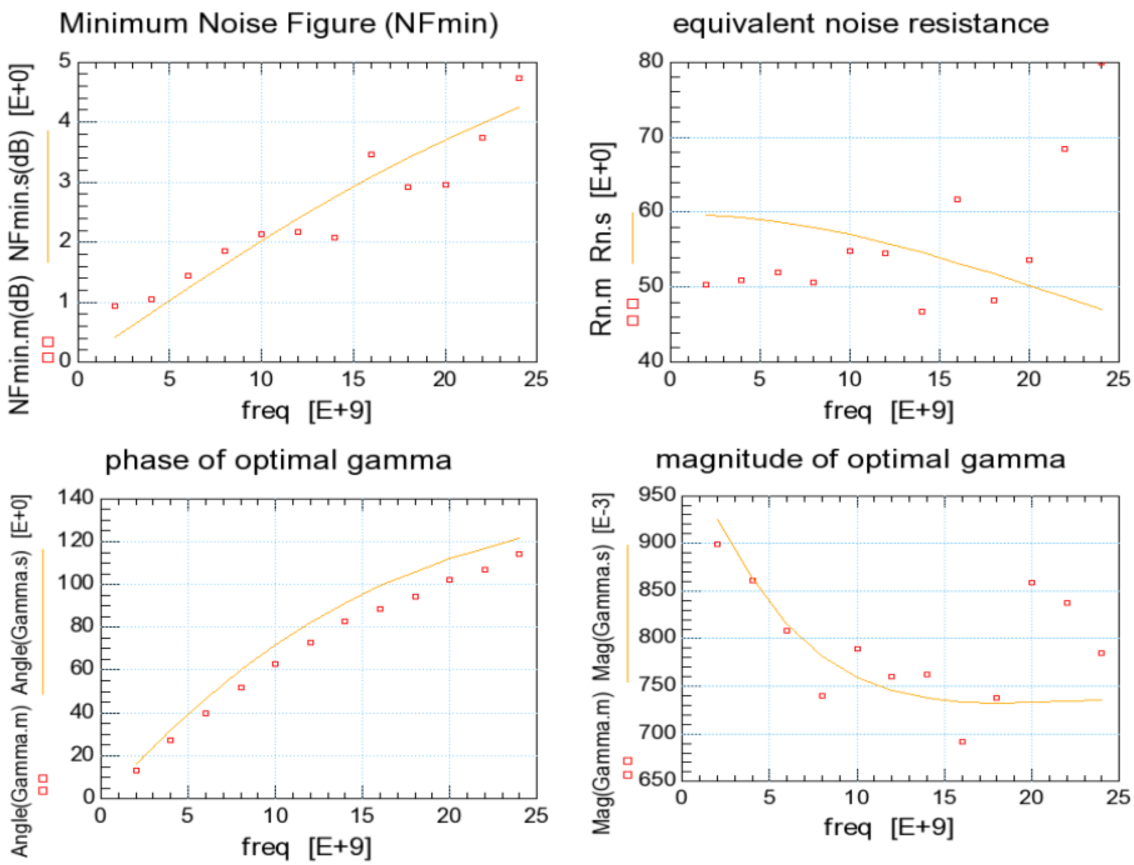


Figure 3-7 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.15um, WF=2.4um, NF=16, m=4 @ Vd=-0.4, Vg=-0.6V

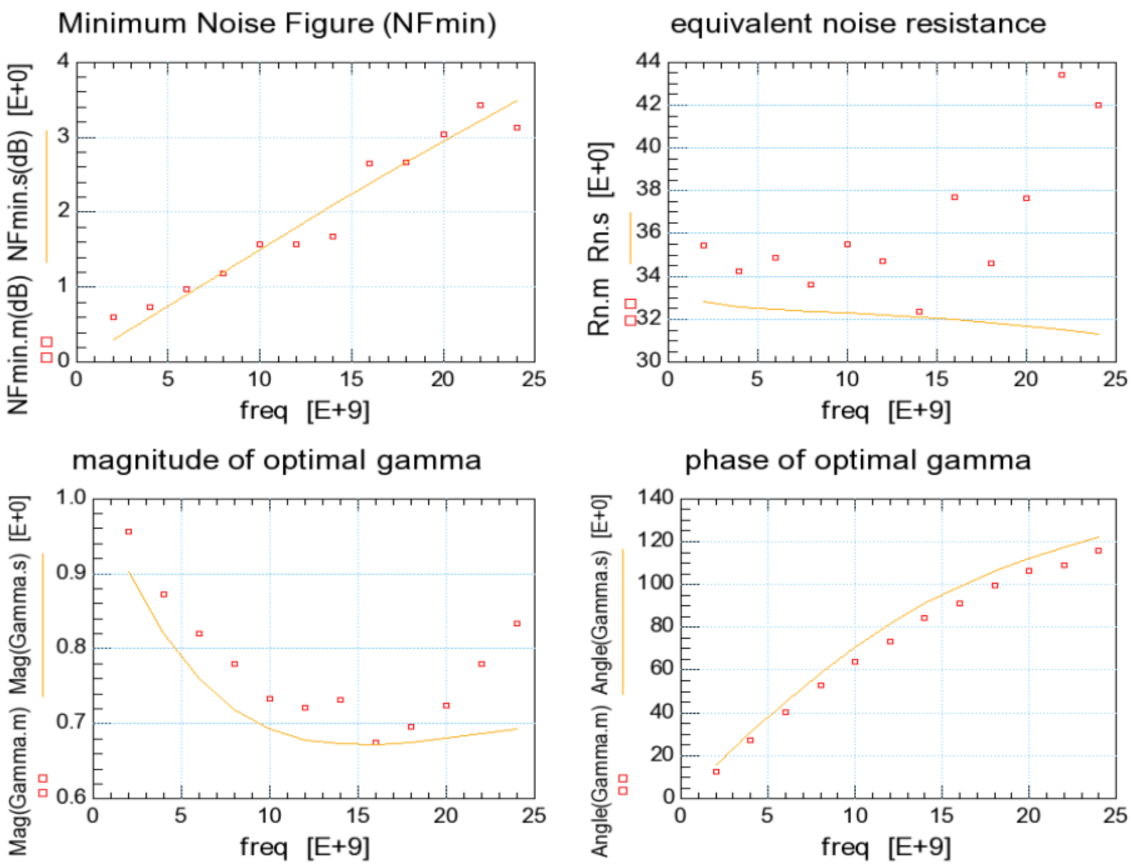


Figure 3-8 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.15um, WF=2.4um, NF=16, m=4 @ Vd=-1.2, Vg=-1.2V

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

4.2 SPECTRE Warning Message

Table 4-2 SPECTRE warning message

NO.	Warning Message	Explanation
1.	None	None

4.3 ELDO Warning Message

Table 4-3 ELDO warning message

NO.	Warning Message	Explanation
1.	None	None